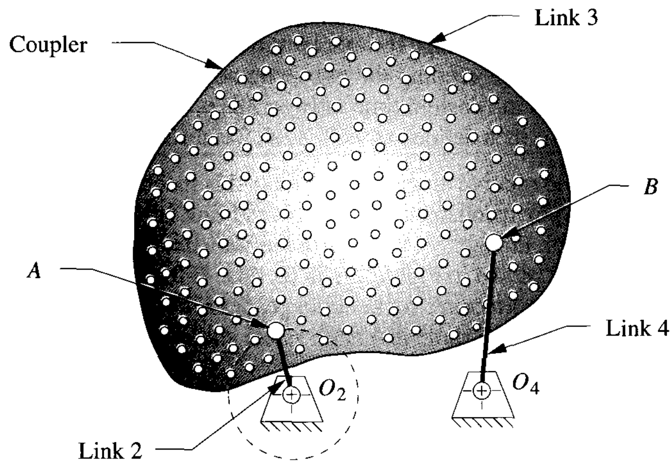
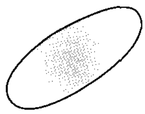
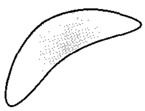
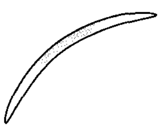


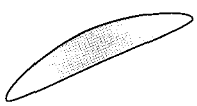
FIGURE 3-15
The fourbar coupler extended to include a large number of coupler points


(a) Pseudo ellipse


(b) Kidney bean


(c) Banana


(d) Crescent


(e) Single straight

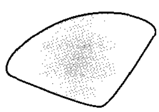

(f) Double straight

FIGURE 3-16 Part 1
A "Cursory Catalog" of coupler curve shapes

in the next section. As we shall see, approximate straight-line motions, dwell motions, and more complicated symphonies of timed motions are available from even the simple fourbar linkage and its infinite variety of often surprising coupler curve motions.

FOURBAR COUPLER CURVES come in a variety of shapes which can be crudely categorized as shown in Figure 3-16. There is an infinite range of variation between these generalized shapes. Some features of interest are the curve's double points, ones that have two tangents. They occur in two types, the cusp and the crunode. A cusp is *a sharp point on the curve which has the useful property of instantaneous zero velocity.* The simplest example of a curve with a cusp is the cycloid curve which is generated by a point on the rim of a wheel rotating on a flat surface. When the point touches the surface, it has the same (zero) velocity as all points on the stationary surface, provided there is pure rolling and no slip between the elements. Anything attached to a cusp point will come smoothly to a stop along one path and then accelerate smoothly away from that point on a different path. The cusp's feature of zero velocity has value in such applications as transporting, stamping and feeding processes. Note that *the acceleration at the cusp is not zero.* A crunode *creates a multiloop curve which has double points at the crossovers.* The two slopes (tangents) at a crunode give the point two different velocities, neither of which is zero in contrast to the cusp. In general, a fourbar coupler curve can have up to three real double points* which may be a combination of cusps and crunodes as can be seen in Figure 3-16.

The *Hrones and Nelson* (H&N) atlas of fourbar coupler curves [8a] is a useful reference which can provide the designer with a starting point for further design and

* Actually, the fourbar coupler curve has 9 double points of which 6 are usually imaginary. However, Fichter and Hunt [8b] point out that some unique configurations of the fourbar linkage (i.e., rhombus parallelograms and those close to this configuration) can have up to 6 real double points which they denote as comprising 3 "proper" and 3 "improper" real double points. For non-special-case Grashof fourbar linkages with minimum transmission angles suitable for engineering applications, only the 3 "proper" double points will appear.

analysis. It contains about 7000 coupler curves and defines the linkage geometry for each of its Grashof crank-rocker linkages. Figure 3-17a reproduces a page from this book. The H&N atlas is logically arranged, with all linkages defined by their link ratios, based on a unit length crank. The coupler is shown as a matrix of fifty coupler points for each linkage geometry, arranged ten to a page. Thus each linkage geometry occupies five pages. Each page contains a schematic "key" in the upper right comer which defines the link ratios.

Figure 3-17b shows a "fleshed out" linkage drawn on top of the H&N atlas page to illustrate its relationship to the atlas information. The double circles in Figure 3-17a define the fixed pivots. The crank is always of unit length. The ratios of the other link lengths to the crank are given on each page. The actual link lengths can be scaled up or down to suit your package constraints and this will affect the size but not the shape of the coupler curve. Anyone of the ten coupler points shown can be used by incorporating it into a triangular coupler link. The location of the chosen coupler point can be scaled from the atlas and is defined within the coupler by the position vector R whose constant angle ϕ is measured with respect to the line of centers of the coupler. The H&N coupler curves are shown as dashed lines. Each dash station represents **five degrees** of crank rotation. So, for an assumed constant crank velocity, the dash spacing is proportional to path velocity. The changes in velocity and the quick-return nature of the coupler path motion can be clearly seen from the dash spacing.

One can peruse this linkage atlas resource and find an approximate solution to virtually any path generation problem. Then one can take the tentative solution from the atlas to a CAE resource such as the FOURBAR program or other package such as *Working Model* * and further refine the design, based on the complete analysis of positions, velocities, and accelerations provided by the program. The only data needed for the FOURBAR program are the four link lengths and the location of the chosen coupler point with respect to the line of centers of the coupler link as shown in Figure 3-17. These parameters can be changed within program FOURBAR to alter and refine the design. Input the file F03-17bAbr to program FOURBAR to animate the linkage shown in that figure.

An example of an application of a fourbar linkage to a practical problem is shown in Figure 3-18 which is a movie camera (or projector) film advance mechanism. Point O_2 is the crank pivot which is motor driven at constant speed. Point O_4 is the rocker pivot, and points *A* and *B* are the moving pivots. Points *A*, *B*, and C define the coupler where C is the coupler point of interest. A movie is really a series of still pictures, each "frame" of which is projected for a small fraction of a second on the screen. Between each picture, the film must be moved very quickly from one frame to the next while the shutter is closed to blank the screen. The whole cycle takes only 1/24 of a second. The human eye's response time is too slow to notice the flicker associated with this discontinuous stream of still pictures, so it appears to us to be a continuum of changing images.

The linkage shown in Figure 3-18 is cleverly designed to provide the required motion. A hook is cut into the coupler of this fourbar Grashof crank-rocker at point C which generates the coupler curve shown. The hook will enter one of the sprocket holes in the **film** as it passes point F_1, Notice that the direction of motion of the hook at that point is nearly perpendicular to the film, so it enters the sprocket hole cleanly. It then turns abruptly downward and follows a crudely approximate straight line as it rapidly pulls the film downward to the next frame. The film is separately guided in a straight track called

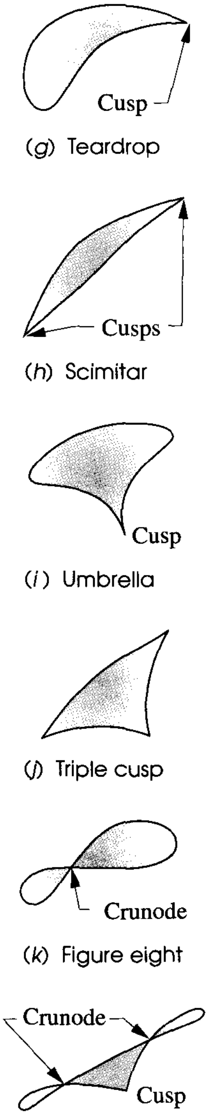
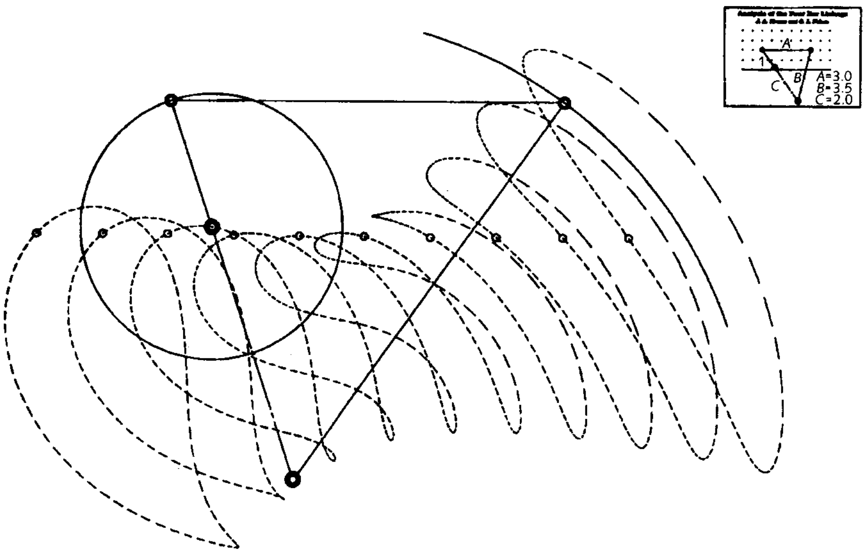


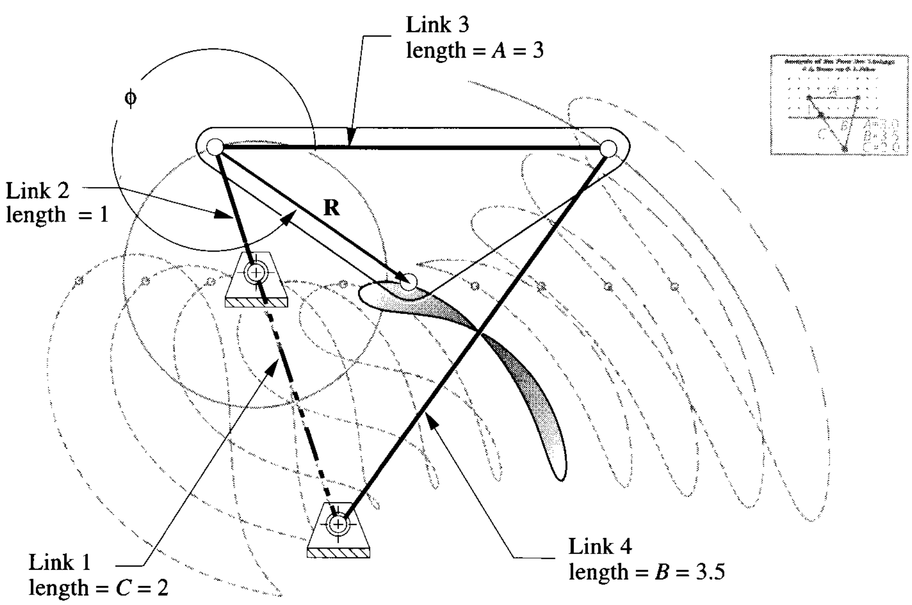
FIGURE 3-16 Part 2
A "Cursory Catalog" of coupler curve shapes

* Included on CD-ROM with this book.



(*a*) A page from the Hrones and Nelson atlas of fourbar coupler curves *

Hrones, J. A., and G. L. Nelson (1951). Analysis of the Fourbar Linkage. MIT Technology Press, Cambridge, MA. Reprinted with permission.



(*b*) Creating the linkage from the information in the atlas

FIGURE 3-17

Selecting a coupler curve and constructing the linkage from the Hrones and Nelson atlas

* The Hrones and Nelson atlas is long out of print but may be available from University Microfilms, Ann Arbor, MI. Also, tbe *Atlas of Linkage Design and Analysis Vall: The Four Bar Linkage* similar to tbe H&N atlas, has been recently published and is available from Saltire Software, 9725 SW Gemini Drive, Beaverton, OR 97005, (800) 659-1874.

the "gate." The shutter (driven by another linkage from the same driveshaft at O2) is closed during this interval of film motion, blanking the screen. At point F_2 there is a cusp on the coupler curve which causes the hook to decelerate smoothly to zero velocity in the vertical direction, and then as smoothly accelerate up and out of the sprocket hole. The abrupt transition of direction at the cusp allows the hook to back out of the hole without jarring the film, which would make the image jump on the screen as the shutter opens. The rest of the coupler curve motion is essentially "wasting time" as it proceeds up the back side, to be ready to enter the film again to repeat the process. Input the file F03-18.4br to program FOURBAR to animate the linkage shown in that figure.

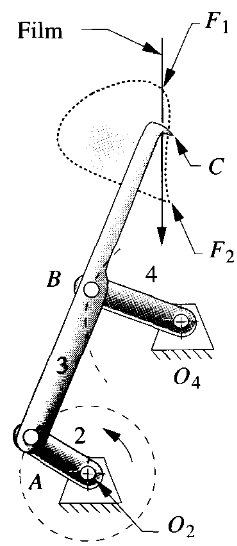


FIGURE 3-18

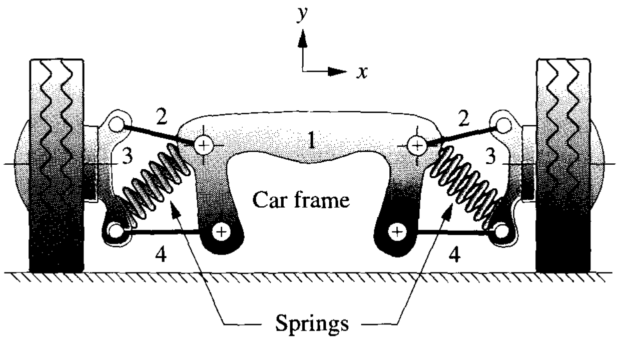
Movie camera film-advance mechanism. *From Die Wissenschaftliche und Angenwandte Photographie, Michel, Kurt,(ed.). (1955). Vol. 3, Harold Weise, Die Kinematographische Kamera, p. 202, Springer Verlag, OHG, Vienna.* (Input the file F03-18.4br to program FOURBAR to animate this linkage.)

Some advantages of using this type of device for this application are that it is very simple and inexpensive (only four links, one of which is the frame of the camera), is extremely reliable, has low friction if good bearings are used at the pivots, and can be reliably timed with the other events in the overall camera mechanism through common shafting from a single motor. There are a myriad of other examples of fourbar coupler curves used in machines and mechanisms of all kinds.

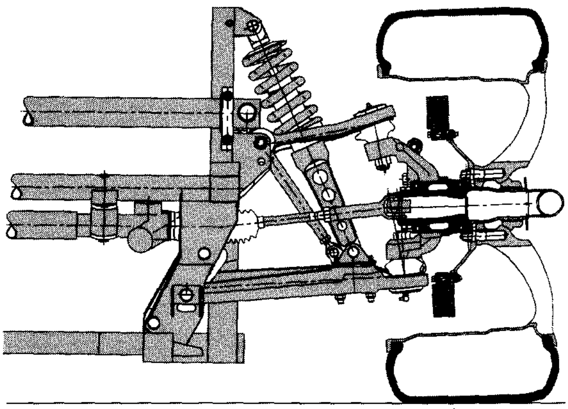
One other example of a very different application is that of the automobile suspension (Figure 3-19). Typically, the up and down motions of the car's wheels are controlled by some combination of planar fourbar linkages, arranged in duplicate to provide three-dimensional control as described in Section 3.2. Only a few manufacturers currently use a true spatial linkage in which the links are not arranged in parallel planes. In all cases the wheel assembly is attached to the coupler of the linkage assembly, and its motion is along a set of coupler curves. The orientation of the wheel is also of concern in this case, so this is not strictly a path generation problem. By designing the linkage to control the paths of multiple points on the wheel (tire contact patch, wheel center, etc.-all of which are points on the same coupler link extended), motion generation is achieved as the coupler has complex motion. Figure 3-19a and b shows parallel planar fourbar linkages suspending the wheels. The coupler curve of the wheel center is nearly a straight line over the small vertical displacement required. This is desirable as the idea is to keep the tire perpendicular to the ground for best traction under all cornering and attitude changes of the car body. This is an application in which a non-Grashof linkage is perfectly acceptable, as full rotation of the wheel in this plane might have some undesirable results and surprise the driver. Limit stops are of course provided to prevent such behavior, so even a Grashof linkage could be used. The springs support the weight of the vehicle and provide a fifth, variable-length "force link" which stabilizes the mechanism as was described in Section 2.14 (p. 54). The function of the fourbar linkage is solely to guide and control the wheel motions. Figure 3-19c shows a true spatial linkage of seven links (including frame and wheel) and nine joints (some of which are ball-and-socket joints) used to control the motion of the rear wheel. These links do not move in parallel planes but rather control the three-dimensional motion of the coupler which carries the wheel assembly.

SYMMETRICAL FOURBAR COUPLER CURVES When a fourbar linkage's geometry is such that the coupler and rocker are the same length pin-to-pin, all coupler points that lie on a circle centered on the coupler-rocker joint with radius equal to the coupler length will generate symmetrical coupler curves. Figure 3-20 shows such a linkage, its symmetrical coupler curve, and the locus of all points that will give symmetrical curves. Using the notation of that figure, the criterion for coupler curve symmetry can be stated as:

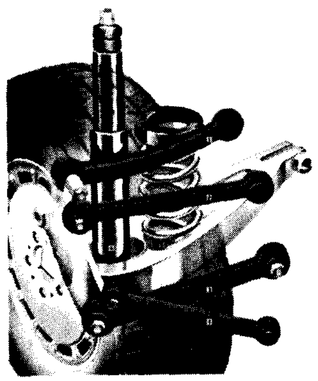
$$AB = O_4B = BP \tag{3.4}$$



(*a*) Fourbar planar linkages are duplicated in parallel planes, displaced in the *z* direction, behind the links shown



(*b*) Parallel-planar linkage used to control Viper wheel motion (*Courtesy of Chrysler Corporation*)



(*c*) Multi-link true spatial linkage used to control rear wheel motion (*Courtesy of Mercedes-Benz of North America Inc.*)

FIGURE 3-19

Linkages used in automotive chassis suspensions

A linkage for which equation 3.4 is true is referred to as a **symmetrical fourbar linkage**. The axis of symmetry of the coupler curve is the line O_4P drawn when the crank O_2A and the ground link O_2O_4 are colinear-extended (i.e., $\theta_2 = 180°$). Symmetrical coupler curves prove to be quite useful as we shall see in the next several sections. Some give good approximations to circular arcs and others give very good approximations to straight lines (over a portion of the coupler curve).

In the general case, nine parameters are needed to define the geometry of a **nonsymmetrical fourbar linkage** with one coupler point.* We can reduce this to five as follows. Three parameters can be eliminated by fixing the location and orientation of the ground link. The four link lengths can be reduced to three parameters by normalizing three link lengths to the fourth. The shortest link (the crank if a Grashof linkage) is usually taken as the reference link, and three link ratios are formed as L_1 / L_2, L_3 / L_2, L_4 / L_2, where

* The nine independent parameters of a fourbar linkage are: four link lengths, two coordinates of the coupler point with respect to the coupler link, and three parameters that define the location and orientation of the fixed link in the global coordinate system.

L_1 = ground, L_2 = crank, L_3 = coupler, and L_4 = rocker length as shown in Figure 3-20. Two parameters are needed to locate the coupler point: the distance from a convenient reference point on the coupler (either B or A in Figure 3-20) to the coupler point P, and the angle that the line BP (or AP) makes with the line of centers of the coupler AB (either δ or γ). Thus, with a defined ground link, five parameters that will define the geometry of a nonsymmetrical fourbar linkage (using point B as the reference in link 3 and the labels of Figure 3-20) are: L_1 / L_2, L_3 / L_2, L_4 / L_2, BP / L_2, and γ. Note that multiplying these parameters by a scaling factor will change the size of the linkage and its coupler curve but will not change the coupler curve's shape.

A **symmetrical fourbar linkage** with a defined ground link needs only *three parameters* to define its geometry because three of the five nonsymmetrical parameters are now equal per equation 3.4: $L_3 / L_2 = L_4 / L_2 = BP / L_2$. Three possible parameters to define the geometry of a symmetrical fourbar linkage in combination with equation 3.4 are then: L_1 / L_2, L_3 / L_2, and γ. Having only three parameters to deal with rather than five greatly simplifies an analysis of the behavior of the coupler curve shape when the linkage geometry is varied. Other relationships for the isosceles-triangle coupler are shown in Figure 3-20. Length AP and angle δ are needed for input of the linkage geometry to program FOURBAR.

Kota [9] did an extensive study of the characteristics of coupler curves of symmetrical fourbar linkages and mapped coupler curve shape as a function of the three linkage parameters defined above. He defined a three-dimensional design space to map the coupler curve shape. Figure 3-21 shows two orthogonal plane sections taken through this design space for particular values of link ratios,* and Figure 3-22 shows a schematic of

$$\delta = \frac{180° - \gamma}{2}$$

$$AP = 2(AB)\cos\delta$$

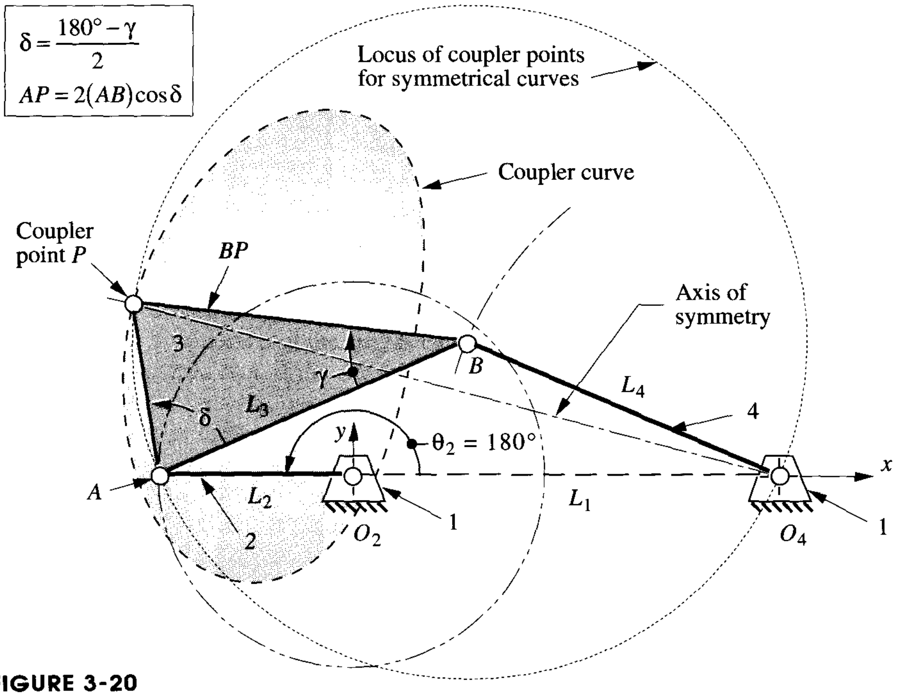
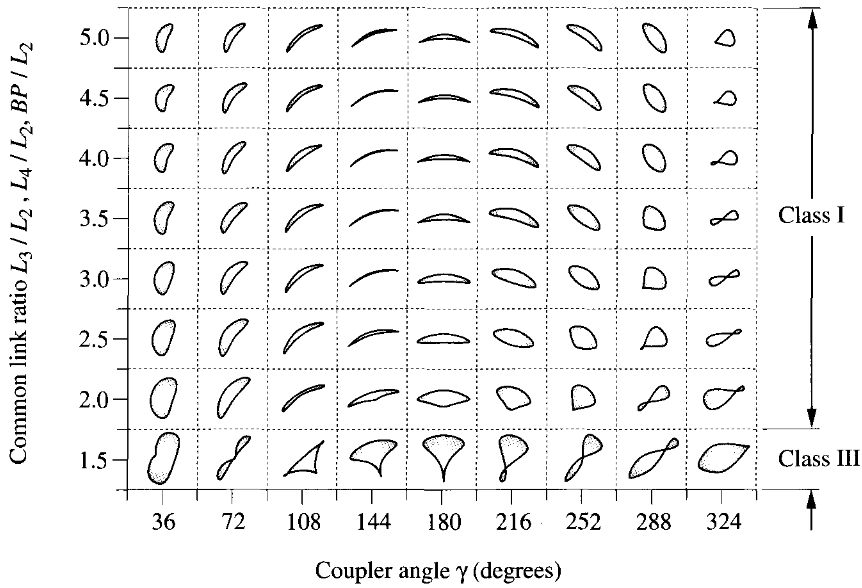


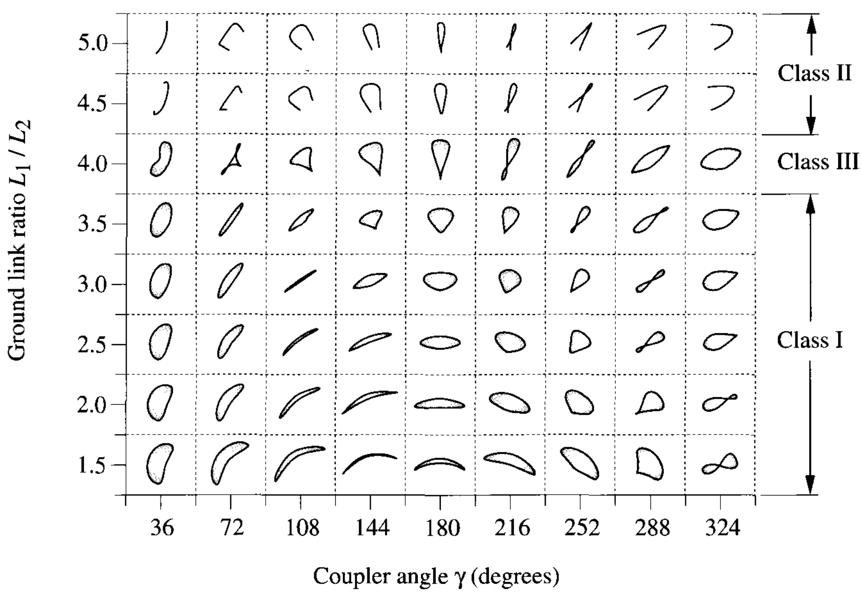
FIGURE 3-20

A fourbar linkage with a symmetrical coupler curve

* Adapted from materials provided by Professor Sridhar Kota, University of Michigan.



(*a*) Variation of coupler curve shape with common link ratio and coupler angle for a ground link ratio $L_1 / L_2 = 2.0$



(*b*) Variation of coupler curve shape with ground link ratio and coupler angle for a common link ratio $L_3 / L_2 = L_4 / L_2 = BP / L_2 = 2.5$

FIGURE 3-21

Coupler curve shapes of symmetrical fourbar linkages *Adapted from reference (9)*

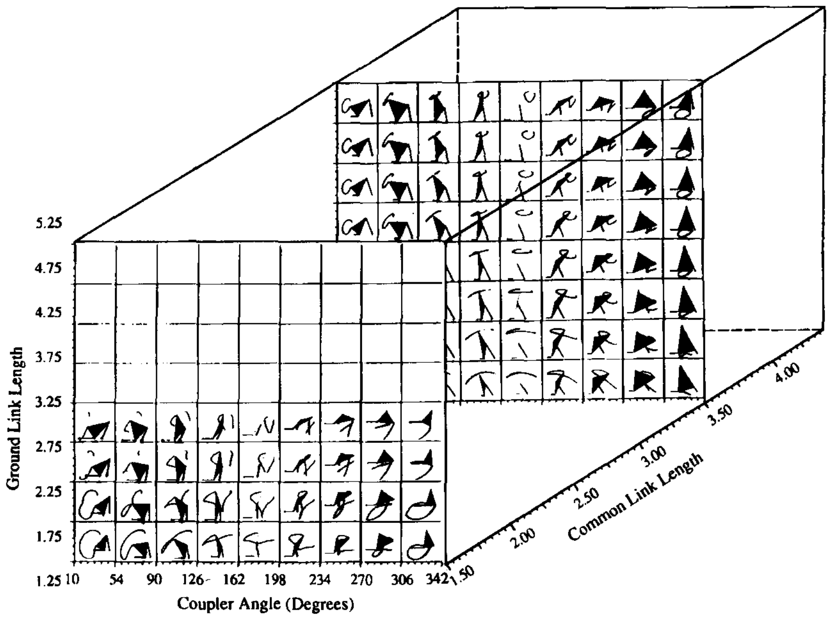


FIGURE 3-22

A three-dimensional map of coupler curve shapes of symmetrical fourbar linkages (9)

the design space. Though the two cross sections of Figure 3-21 show only a small fraction of the information in the 3-D design space of Figure 3-22, they nevertheless give a sense of the way that variation of the three linkage parameters affects the coupler curve shape. Used in combination with a linkage design tool such as program FOURBAR, these design charts can help guide the designer in choosing suitable values for the linkage parameters to achieve a desired path motion.

GEARED FIVEBAR COUPLER CURVES (Figure 3-23) are more complex than the fourbar variety. Because there are three additional, independent design variables in the geared fivebar compared to the fourbar (an additional link ratio, the gear ratio, and the phase angle between the gears), the coupler curves can be of higher degree than those of the fourbar. This means that the curves can be more convoluted, having more cusps and crunodes (loops). In fact, if the gear ratio used is noninteger, the input link will have to make a number of revolutions equal to the factor necessary to make the ratio an integer before the coupler curve pattern will repeat. The Zhang, Norton, Hammond (ZNH) *Atlas of Geared FiveBar Mechanisms* (GFBM) [10] shows typical coupler curves for these linkages limited to symmetrical geometry (e.g., link 2 = link 5 and link 3 = link 4) and gear ratios of ±1 and ±2. A page from the ZNH atlas is reproduced in Figure 3-23. Additional pages are in Appendix E. Each page shows the family of coupler curves obtained by variation of the phase angle for a particular set of link ratios and gear ratio. A key in the upper right corner of each page defines the ratios: α = link 3 / link 2, β = link 1 / link 2, λ = gear 5 / gear 2. Symmetry defines links 4 and 5 as noted above. The phase angle ϕ is noted on the axes drawn at each coupler curve and can be seen to have a significant effect on the resulting coupler curve shape.

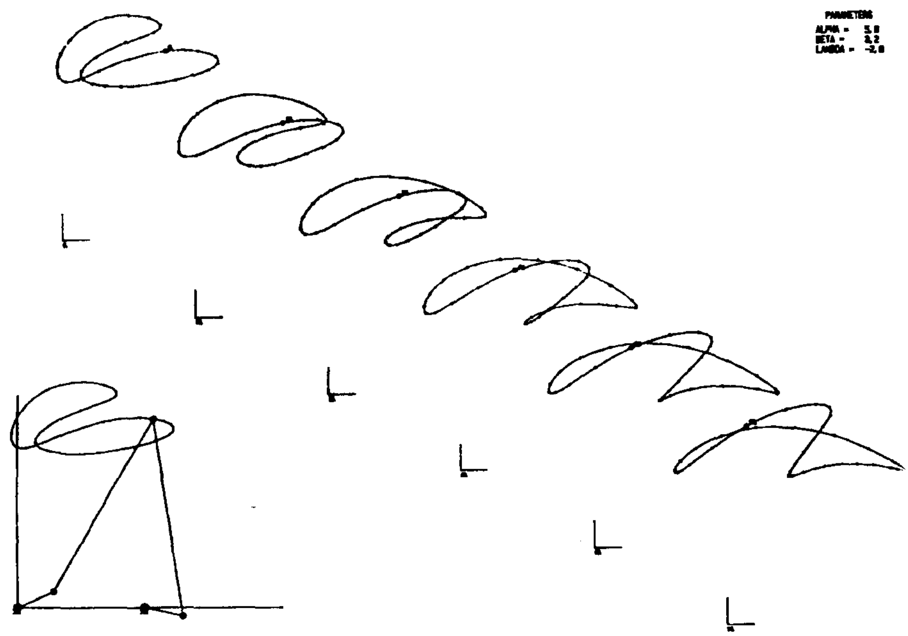


FIGURE 3-23

A page from the Zhang-Norton-Hammond atlas of geared fivebar coupler curves [10]

This reference atlas is intended to be used as a starting point for a geared fivebar linkage design. The link ratios, gear ratio, and phase angle can be input to the program FIVEBAR and then varied to observe the effects on coupler curve shape, velocities, and accelerations. Asymmetry of links can be introduced, and a coupler point location other than the pin joint between links 3 and 4 defined within the FIVEBAR program as well. Note that program FIVEBAR expects the gear ratio to be in the form gear 2 / gear 5 which is the inverse of the ratio λ in the ZNH atlas.

3.7 COGNATES

It sometimes happens that a good solution to a linkage synthesis problem will be found that satisfies path generation constraints but which has the fixed pivots in inappropriate locations for attachment to the available ground plane or frame. **In** such cases, the use of a **cognate** to the linkage may be helpful. The term **cognate** was used by Hartenberg and Denavit [11] to describe *a linkage, of different geometry, which generates the same coupler curve.* Samuel Roberts (1875) and Chebyschev (1878) independently discovered the theorem which now bears their names:

Roberts-Chebyschev Theorem

Three different planar, pin-jointedfourbar linkages will trace identical coupler curves.

Hartenberg and Denavit[1I] presented extensions of this theorem to the slider-crank and the sixbar linkages:

Two different planar slider-crank linkages will trace identical coupler curves.

The coupler-point curve of a planar fourbar linkage is *also described by the joint of a dyad of an appropriate sixbar linkage.*

Figure 3-24a shows a fourbar linkage for which we want to find the two cognates. The first step is to release the fixed pivots D_A and D_B. While holding the coupler stationary, rotate links 2 and 4 into colinearity with the line of centers (A_1B_1) of link 3 as shown in Figure 3-24b. We can now construct lines parallel to all sides of the links in the original linkage to create the Cayley diagram in Figure 3-24c. This schematic arrangement defines the lengths and shapes oflinks 5 through 10 which belong to the cognates. All three fourbars share the original coupler point P and will thus generate the same path motion on their coupler curves.

In order to find the correct location of the fixed pivot D_C from the Cayley diagram, the ends of links 2 and 4 are returned to the original locations of the fixed pivots D_A and D_B as shown in Figure 3-25a. The other links will follow this motion, maintaining the parallelogram relationships between links, and fixed pivot D_C will then be in its proper location on the ground plane. This configuration is called a Roberts diagram-three fourbar linkage cognates which share the same coupler curve.

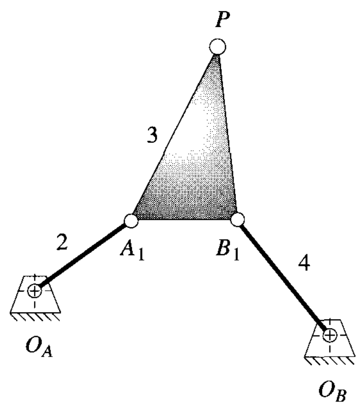
The Roberts diagram can be drawn directly from the original linkage without resort to the Cayley diagram by noting that the parallelograms which form the other cognates are also present in the Roberts diagram and the three couplers are similar triangles. It is also possible to locate fixed pivot D_C directly from the original linkage as shown in Figure 3-25a. Construct a similar triangle to that of the coupler, placing its base (AB) between D_A and D_B. Its vertex will be at D_C.

The ten-link Roberts configuration (Cayley's nine plus the ground) can now be articulated up to any toggle positions, and point P will describe the original coupler path which is the same for all three cognates. Point D_C will not move when the Roberts linkage is articulated, proving that it is a ground pivot. The cognates can be separated as shown in Figure 3-25b and anyone of the three linkages used to generate the same coupler curve. Corresponding links in the cognates will have the same angular velocity as the original mechanism as defined in Figure 3-25.

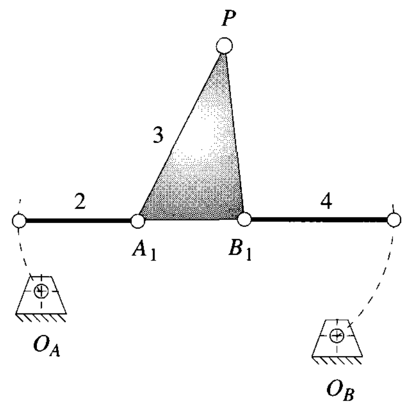
Nolle [12] reports on work by Luck [13] (in German) that defines the character of all fourbar cognates and their transmission angles. If the original linkage is a Grashof crank-rocker, then one cognate will be also, and the other will be a Grashof double rocker. The minimum transmission angle of the crank-rocker cognate will be the same as that of the original crank-rocker. If the original linkage is a Grashof double-crank (drag link), then both cognates will be also and their minimum transmission angles will be the same in pairs that are driven from the same fixed pivot. If the original linkage is a non-Grashof triple-rocker, then both cognates are also triple-rockers.

These findings indicate that cognates of Grashof linkages do not offer improved transmission angles over the original linkage. Their main advantages are the different fixed pivot location and different velocities and accelerations of other points in the linkage. While the coupler path is the same for all cognates, its velocities and accelerations will not generally be the same since each cognate's overall geometry is different.

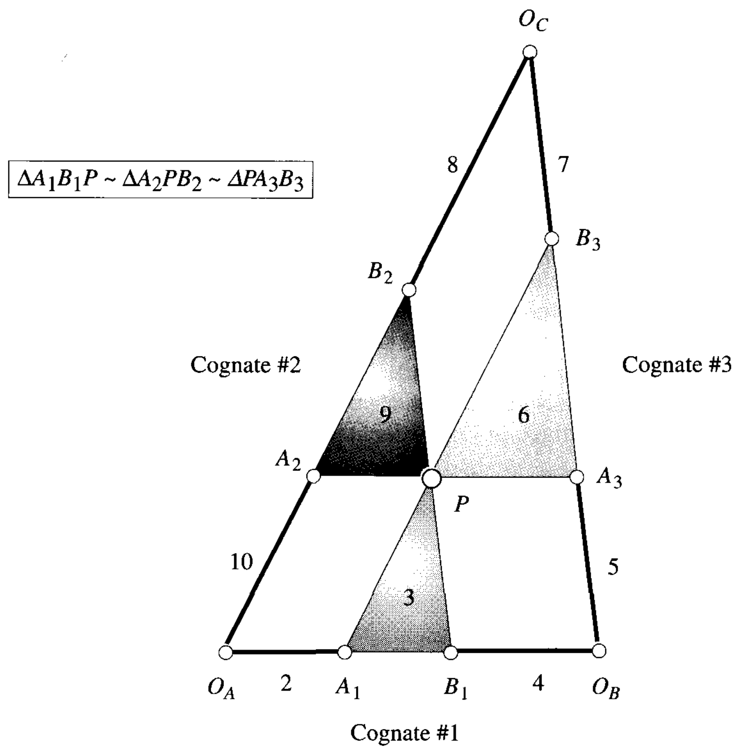
When the coupler point lies on the line of centers of link 3, the Cayley diagram degenerates to a group of colinear lines. A different approach is needed to determine the



(a) Original fourbar linkage (cognate #1)



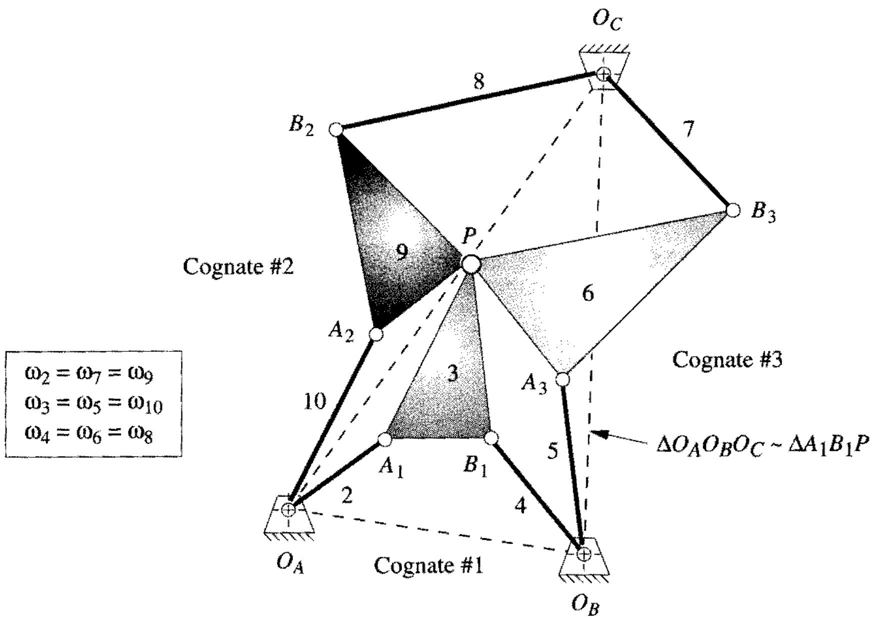
(b) Align links 2 and 4 with coupler



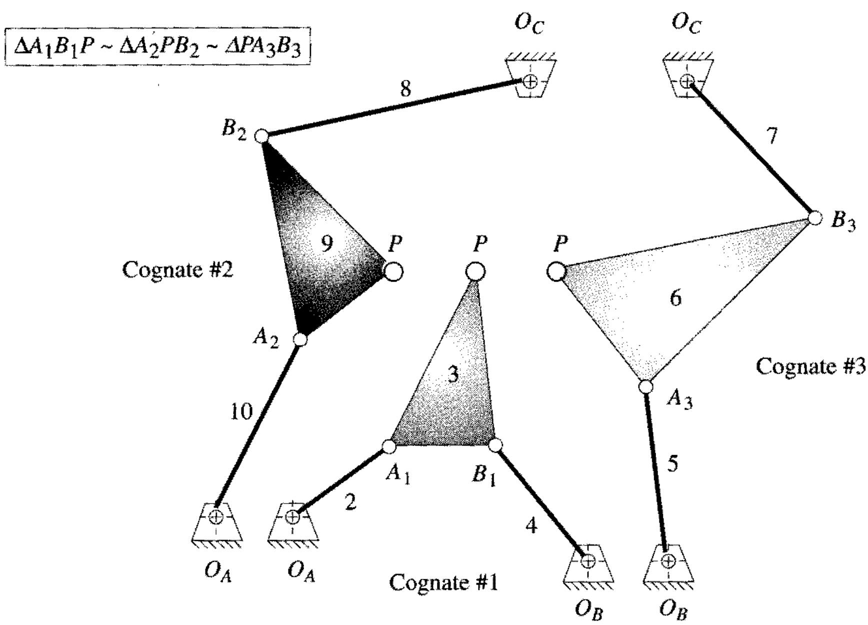
(c) Construct lines parallel to all sides of the original fourbar linkage to create cognates

FIGURE 3-24

Cayley diagram to find cognates of a fourbar linkage



(a) Return links 2 and 4 to their fixed pivots O_A and O_B. Point O_C will assume its proper location



(b) Separate the three cognates. Point P has the same path motion in each cognate.

FIGURE 3-25

Roberts diagram of three fourbar cognates geometry of the cognates. Hartenberg and Denavit [11] give the following set of steps to find the cognates in this case. The notation refers to Figure 3-26.

1 Fixed pivot O_C lies on the line of centers O_AO_B extended and divides it in the same ratio as point P divides AB (i.e., $O_C / O_A = PA / AB$).

2 Line O_AA_2 is parallel to A_1P and A_2P is parallel to O_AA_1, locating A_2.

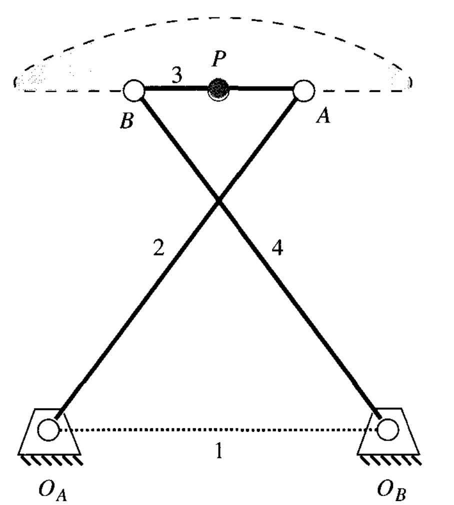
3 Line O_BA_3 is parallel to B_1P and A_3P is parallel to O_BB_1, locating A_3.

4 Joint B_2 divides line A_2P in the same ratio as point P divides AB. This defines the first cognate $O_AA_2B_2O_C$.

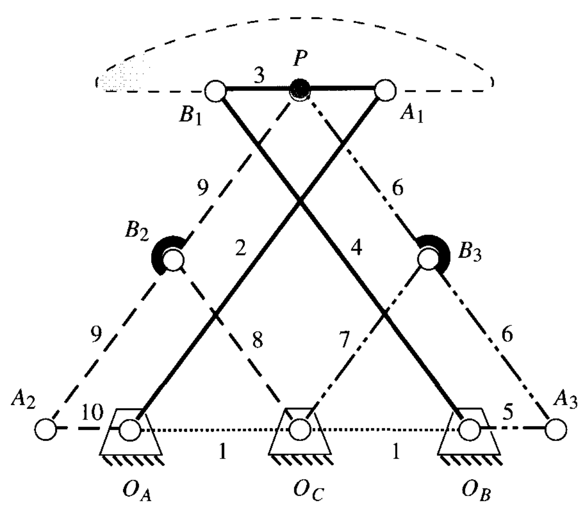
5 Joint B_3 divides line A_3P in the same ratio as point P divides AB. This defines the second cognate $O_BA_3B_3O_C$.

The three linkages can then be separated and each will independently generate the same coupler curve. The example chosen for Figure 3-26 is unusual in that the two cognates of the original linkage are identical, mirror-image twins. These are special linkages and will be discussed further in the next section.

Program FOURBAR will automatically calculate the two cognates for any linkage configuration input to it. The velocities and accelerations of each cognate can then be calculated and compared. The program also draws the Cayley diagram for the set of cognates. Input the file F03-24.4br to program FOURBAR to display the Cayley diagram of Figure 3-24 (p.114). Input the files COGNATE1.4br, COGNATE2.4br, and COGNATE3.4br to animate and view the motion of each cognate shown in Figure 3-25 (p. 115). Their coupler curves (at least those portions that each cognate can reach) will be seen to be identical.



(a) A fourbar linkage and its coupler curve



(b) Cognates of the fourbar linkage

FIGURE 3-26

Finding cognates of a fourbar linkage when its coupler point lies on the line of centers of the coupler

Parallel Motion

It is quite common to want the output link of a mechanism to follow a particular path without any rotation of the link as it moves along the path. Once an appropriate path motion in the form of a coupler curve and its fourbar linkage have been found, a cognate of that linkage provides a convenient means to replicate the coupler path motion and provide curvilinear translation (i.e., no rotation) of a new, output link that follows the coupler path. This is referred to as **parallel motion.** Its design is best described with an example, the result of which will be a Watt's I sixbar linkage* that incorporates the original fourbar and parts of one of its cognates. The method shown is as described in Soni.[14]

EXAMPLE 3-11

Parallel Motion from a Fourbar Linkage Coupler Curve.

Problem: Design a sixbar linkage for parallel motion over a fourbar linkage coupler path.

Solution: (see Figure 3-27)

1 Figure 3-27a shows the chosen Grashof crank-rocker fourbar linkage and its coupler curve. The first step is to create the Roberts diagram and find its cognates as shown in Figure 3-27b. The Roberts linkage can be found directly, without resort to the Cayley diagram, as described above. The fixed center O_C is found by drawing a triangle similar to the coupler triangle A_1B_1P with base O_AO_B.

2 One of a crank-rocker linkage's cognates will also be a crank-rocker (here cognate #3) and the other is a Grashof double-rocker (here cognate #2). Discard the double-rocker, keeping the links numbered 2, 3, 4, 5, 6, and 7 in Figure 3-27b. Note that links 2 and 7 are the two cranks, and both have the same angular velocity. The strategy is to coalesce these two cranks on a common center (O_A) and then combine them into a single link.

3 Draw the line qq parallel to line O_AO_C and through point O_B as shown in Figure 3-27c.

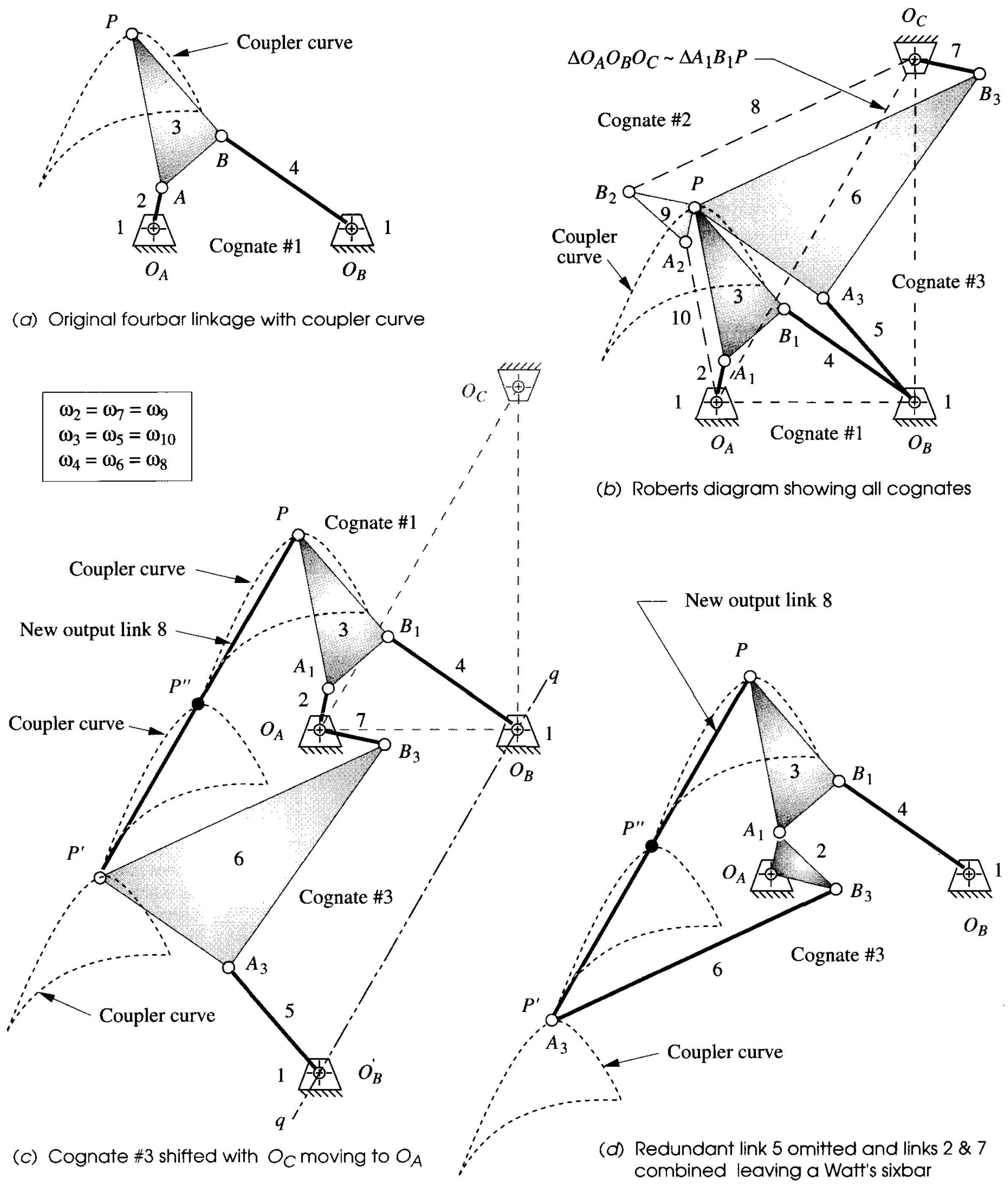
4 Without allowing links 5, 6, and 7 to rotate, slide them as an assembly along lines O_AO_C and qq until the free end of link 7 is at point O_A. The free end of link 5 will then be at point O_B' and point P on link 6 will be at P'.

5 Add a new link of length O_AO_C between P and P'. This is the *new output link* 8 and all points on it describe the original coupler curve as depicted at points P, P', and P'' in Figure 3-27c.

6 The mechanism in Figure 3-27c has 8 links, 10 revolute joints, and one DOF. When driven by **either** crank 2 or 7, all points on link 8 will duplicate the coupler curve of point P.

7 This is an *overclosed linkage* with redundant links. Because links 2 and 7 have the same angular velocity, they can be joined into one link as shown in Figure 3-27d. Then link 5 can be removed and link 6 reduced to a binary link supported and constrained as part of the loop 2, 6, 8, 3. The resulting mechanism is a Watt-I sixbar (see Figure 2-14, p. 45) with the links numbered 1, 2, 3, 4, 6, and 8. Link 8 is in *curvilinear translation* and follows the coupler path of the original point P.

* Another common method used to obtain parallel motion is to duplicate the same linkage (i.e., the identical cognate), connect them with a parallelogram loop, and remove two redundant links. This results in an eight-link mechanism. See Figure P3-7 on p. 135 for an example of such a mechanism. The method shown here using a different cognate results in a simpler linkage, but either approach will accomplish the desired goal.



(a) Original fourbar linkage with coupler curve

(b) Roberts diagram showing all cognates

(c) Cognate #3 shifted with O_C moving to O_A

(d) Redundant link 5 omitted and links 2 & 7 combined leaving a Watt's sixbar

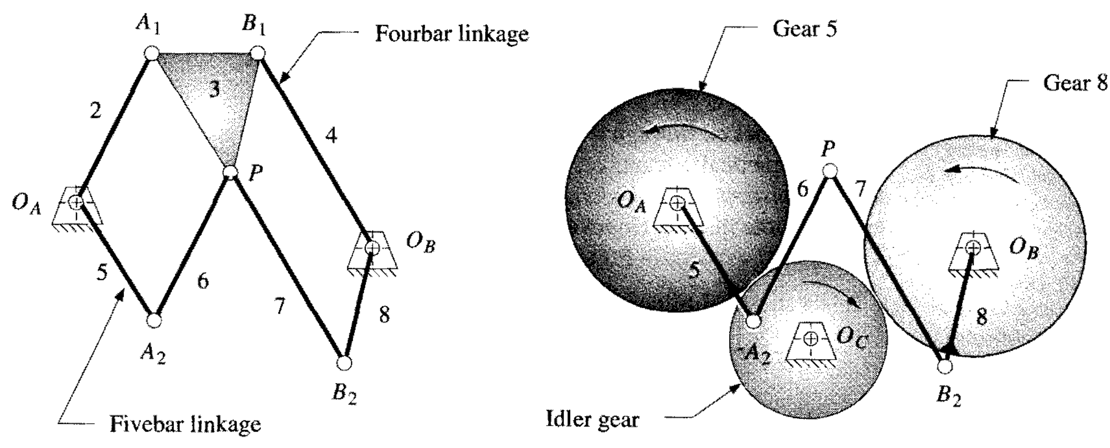
FIGURE 3-27

Method to construct a Watt-I sixbar that replicates a coupler path with curvilinear translation (parallel motion) (14)

Geared Fivebar Cognates of the Fourbar

Chebyschev also discovered that any fourbar coupler curve can be duplicated with a **geared fivebar mechanism whose gear ratio is plus one,** meaning that the gears turn with the same speed and direction. The geared fivebar's link lengths will be different from those of the fourbar but can be determined directly from the fourbar. Figure 3-28a shows the construction method, as described by Hall [15], to obtain the geared fivebar which will give the same coupler curve as a fourbar. The original fourbar is $O_AA_1B_1O_B$ (links 1, 2, 3, 4). The fivebar is $O_AA_2PB_2O_B$ (links 1, 5, 6, 7, 8). The two linkages share only the coupler point P and fixed pivots O_A and O_B. The fivebar is constructed by simply drawing link 6 parallel to link 2, link 7 parallel to link 4, link 5 parallel to A_1P, and link 8 parallel to B_1P.

A three-gear set is needed to couple links 5 and 8 with a ratio of plus one (gear 5 and gear 8 are the same diameter and have the same direction of rotation, due to the idler gear), as shown in Figure 3-28b. Link 5 is attached to gear 5, as is link 8 to gear 8. This construction technique may be applied to each of the three fourbar cognates, yielding three geared fivebars (which may or may not be Grashof). The three fivebar cognates can actually be seen in the Roberts diagram. Note that in the example shown, a non-Grashof triple-rocker fourbar yields a Grashof fivebar, which can be motor driven. This conversion to a GFBM linkage could be an advantage when the "right" coupler curve has been found on a non-Grashof fourbar linkage, but continuous output through the fourbar's toggle positions is needed. Thus we can see that there are at least seven linkages which will generate the same coupler curve, three fourbars, three GFBMs and one or more sixbars.



(a) Construction of equivalent fivebar linkage (b) Resulting geared fivebar linkage

FIGURE 3-28

A geared fivebar linkage cognate of a fourbar linkage

Program FOURBAR calculates the equivalent geared fivebar configuration for any fourbar linkage and will export its data to a disk file that can be opened in program FIVEBAR for analysis. The file F03-28a.4br can be opened in FOURBAR to animate the linkage shown in Figure 3-28a. Then also open the file F03-28b.5br in program FIVEBAR to see the motion of the equivalent geared fivebar linkage. Note that the original fourbar linkage is a triple-rocker, so cannot reach all portions of the coupler curve when driven from one rocker. But, its geared fivebar equivalent linkage can make a full revolution and traverses the entire coupler path. To export a FIVEBAR disk file for the equivalent GFBM of any fourbar linkage from program FOURBAR, use the *Export* selection under the *File* pull-down menu.

3.8 STRAIGHT-LINE MECHANISMS

A very common application of coupler curves is the generation of approximate straight lines. Straight-line linkages have been known and used since the time of James Watt in the 18th century. Many kinematicians such as Watt, Chebyschev, Peaucellier, Kempe, Evans, and Hoeken (as well as others) over a century ago, developed or discovered either approximate or exact straight-line linkages, and their names are associated with those devices to this day.

The first recorded application of a coupler curve to a motion problem is that of **Watt's straight-line linkage,** patented in 1784, and shown in Figure 3-29a. Watt devised his straight-line linkage to guide the long-stroke piston of his steam engine at a time when metal-cutting machinery that could create a long, straight guideway did not yet exist. * This triple-rocker linkage is still used in automobile suspension systems to guide the rear axle up and down in a straight line as well as in many other applications.

Richard Roberts (1789-1864) (not to be confused with Samuel Roberts of the cognates) discovered the **Roberts' straight-line linkage** shown in Figure 3-29b. This is a triple-rocker. **Chebyschev** (1821-1894) also devised a **straight-line linkage**—a Grashof double-rocker—shown in Figure 3-29c.

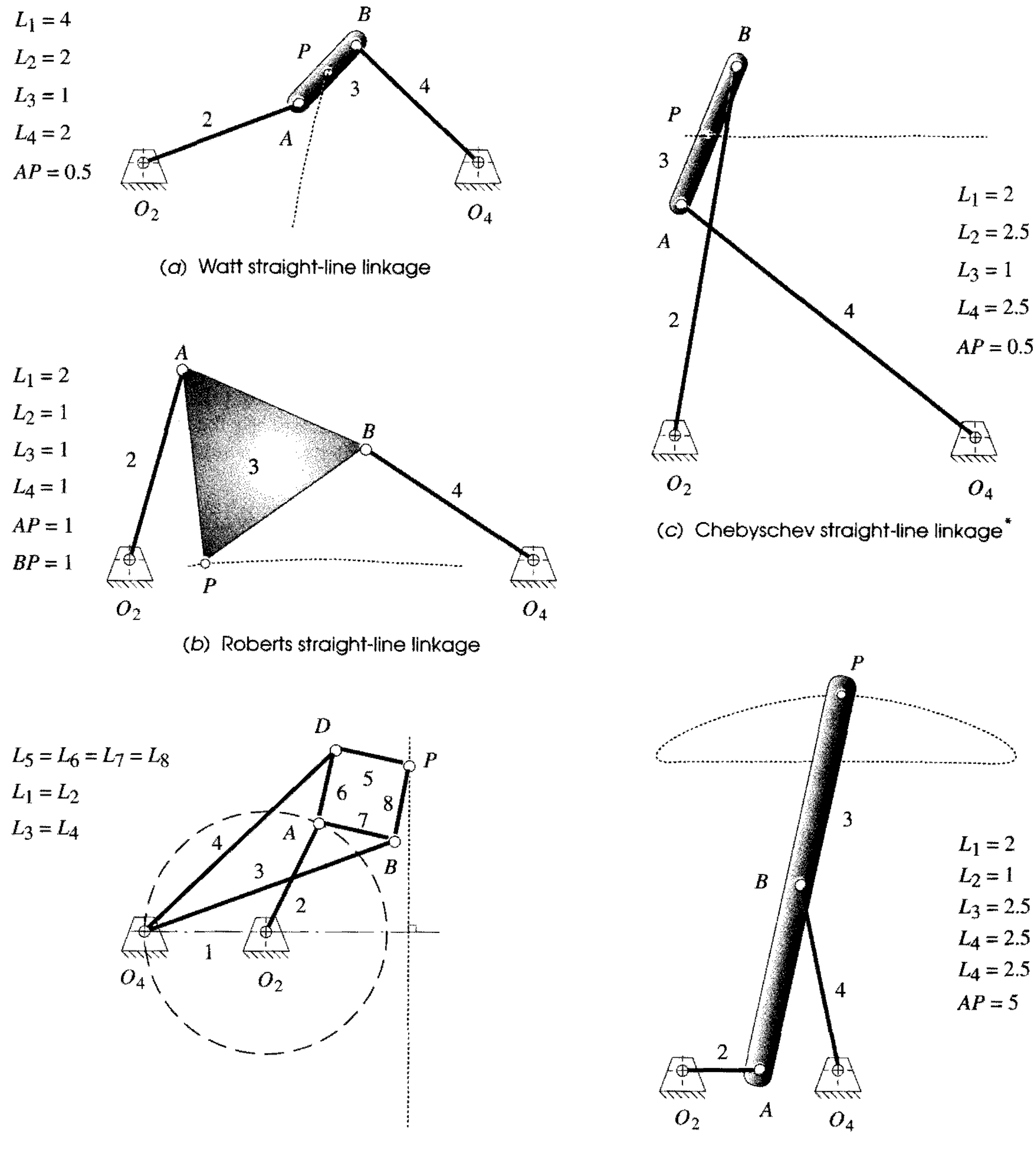
The **Hoeken linkage** [16] in Figure 3-29d is a Grashof crank-rocker, which is a significant practical advantage. In addition, the Hoeken linkage has the feature of very *nearly constant velocity along the center portion of its straight-line motion.* It is interesting to note that the **Hoecken** and **Chebyschev** linkages are cognates of one another. † The cognates shown in Figure 3-26 (p. 116) are the Chebyschev and Hoeken linkages.

These straight-line linkages are provided as built-in examples in program FOURBAR. A quick look in the Hrones and Nelson atlas of coupler curves will reveal a large number of coupler curves with **approximate straight-line** segments. They are quite common.

To generate an **exact straight line** with only pin joints requires more than four links. At least six links and seven pin joints are needed to generate an exact straight line with a pure revolute-jointed linkage, i.e., a Watt's or Stephenson's sixbar. A geared fivebar mechanism, with a gear ratio of -1 and a phase angle of π radians, will generate an exact straight line at the joint between links 3 and 4. But this linkage is merely a transformed Watt's sixbar obtained by replacing one binary link with a higher joint in the form of a gear pair. This geared fivebar's straight-line motion can be seen by reading the file STRAIGHT.5BR into program FIVEBAR, calculating and animating the linkage.

* In Watt's time, straight-line motion was dubbed "parallel motion" though we use that term somewhat differently now. James Watt is reported to have told his son "*Though I am not over anxious after fame, yet I am more proud of the parallel motion than of any other mechanical invention I have made.*" Quoted in Muirhead, J. P. (1854). *The Origin and Progress of the Mechanical Inventions of James Watt*, Vol. 3, London, p. 89.

† Hain [17] (1967) cites the Hoeken reference [16] (1926) for this linkage. Nolle [18] (1974) shows the Hoeken mechanism but refers to it as a Chebyschev crank-rocker without noting its cognate relationship to the Chebyschev double-rocker, which he also shows. It is certainly conceivable that Chebyschev, as one of the creators of the theorem of cognate linkages, would have discovered the "Hoeken" cognate of his own double-rocker. However, this author has been unable to find any mention of its genesis in the English literature other than the ones cited here.



(a) Watt straight-line linkage

(b) Roberts straight-line linkage

(c) Chebyschev straight-line linkage*

(e) Peaucellier exact straight-line linkage

(d) Hoeken straight-line linkage

FIGURE 3-29

Some common and classic approximate, and one exact, straight-line linkages

* The link ratios of the Chebyschev straight-line linkage have been reported differently by various authors. The ratios used here are those first reported (in English) by Kempe (1877). But Kennedy (1893) describes the same linkage, reportedly "as Chebyschev demonstrated it at the Vienna Exhibition of 1893" as having the link ratios 1, 3.25, 2.5, 3.25. We will assume the earliest reference by Kempe to be correct as listed in the figure.

Peaucellier * (1864) discovered an exact straight-line mechanism of eight bars and six pins, shown in Figure 3-2ge. Links 5, 6, 7, 8 form a rhombus of convenient size. Links 3 and 4 can be any convenient but equal lengths. When *OZ04* exactly equals *OzA,* point C generates an *arc of infinite radius,* i.e., an exact straight line. By moving the pivot Oz left or right from the position shown, changing only the length of link 1, this mechanism *will generate true circle arcs with radii much larger than the link lengths.*

Designing Optimum Straight-Line Fourbar Linkages

Given the fact that an exact straight line can be generated with six or more links using only revolute joints, why use a fourbar approximate straight-line linkage at all? One reason is the desire for simplicity in machine design. The pin-jointed fourbar is the simplest possible one-DOF mechanism. Another reason is that a very good approximation to a true straight line can be obtained with just four links, and this is often "good enough" for the needs of the machine being designed. Manufacturing tolerances will, after all, cause any mechanism's performance to be less than ideal. As the number of links and joints increases, the probability that an exact-straight-line mechanism will deliver its theoretical performance in practice is obviously reduced.

There is a real need for straight-line motions in machinery of all kinds, especially in automated production machinery. Many consumer products such as cameras, film, toiletries, razors, and bottles are manufactured, decorated, or assembled on sophisticated and complicated machines that contain a myriad of linkages and cam-follower systems. Traditionally, most of this kind of production equipment has been of the intermittent-motion variety. This means that the product is carried through the machine on a linear or rotary conveyor that stops for any operation to be done on the product, and then indexes the product to the next work station where it again stops for another operation to be performed. The forces and power required to accelerate and decelerate the large mass of the conveyor (which is independent of, and typically larger than, the mass of the product) severely limit the speeds at which these machines can be run.

Economic considerations continually demand higher production rates, requiring higher speeds or additional, expensive machines. This economic pressure has caused many manufacturers to redesign their assembly equipment for continuous conveyor motion. When the product is in continuous motion in a straight line and at constant velocity, every workhead that operates on the product must be articulated to chase the product and match both its straight-line path and its constant velocity while performing the task. These factors have increased the need for straight-line mechanisms, including ones capable of near-constant velocity over the straight-line path.

A (near) perfect straight-line motion is easily obtained with a fourbar slider-crank mechanism. Ball-bushings (Figure 2-26, p. 57) and hardened ways are available commercially at moderate cost and make this a reasonable, low-friction solution to the straight-line path guidance problem. But, the cost and lubrication problems of a properly guided slider-crank mechanism are still greater than those of a pin-jointed fourbar linkage. Moreover, a crank-slider-block has a velocity profile that is nearly sinusoidal (with some harmonic content) and is far from having constant velocity over any of its motion.

The Hoeken-type linkage offers an 0l?timum combination of straightness and near constant velocity and is a crank-rocker, so it can be motor driven. Its geometry, dimen-

* Peaucellier was a French army captain and military engineer who first proposed his "compas compose" or *compound compass* in 1864 but received no immediate recognition therefor. The British-American mathematician, James Sylvester, reported on it to the *Atheneum Club* in London in 1874. He observed that *"The perfect parallel motion of Peaucellier looks so simple and moves so easily that people who see it at work almost universally express astonishment that it waited so long to be discovered."* A model of the Peaucellier linkage was passed around the table. The famous physicist, Sir William Thomson (later Lord Kelvin), refused to relinquish it, declaring *"No. I have not had nearly enough of it-it is the most beautiful thing I have ever seen in my life."* Source: Strandh, S. (1979). *A History of the Machine.* A&W Publishers: New York, p. 67.

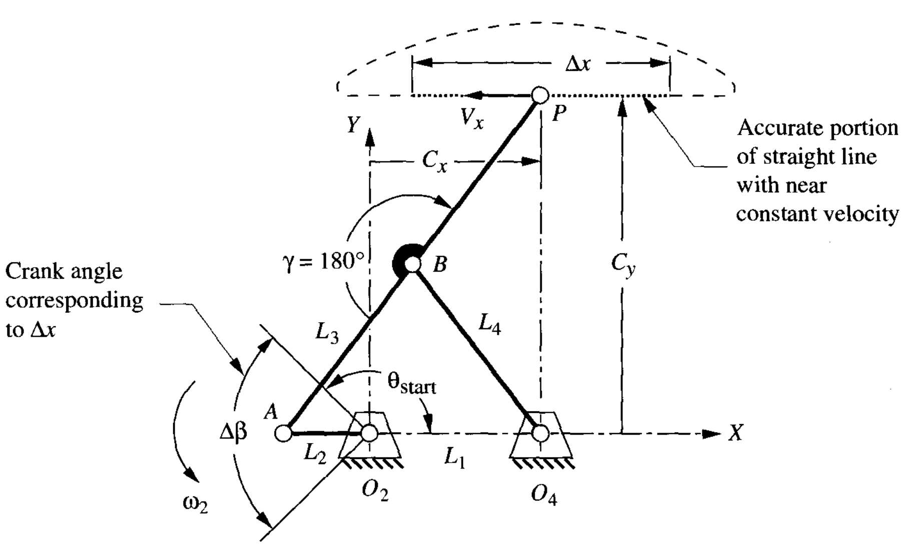


FIGURE 3-30

Hoekens linkage geometry. Linkage shown with P at center of straight-line portion of path

sions, and coupler path are shown in Figure 3-30. This is a symmetrical fourbar linkage. Since the angle γ of line BP is specified and $L_3 = L_4 = BP$, only two link ratios are needed to define its geometry, say L_1 / L_2 and L_3 / L_2. If the crank L_2 is driven at constant angular velocity ω_2, the linear velocity V_x along the straight line portion Δx of the coupler path will be very close to constant over a significant portion of crank rotation $\Delta\beta$.

A study was done to determine the errors in straightness and constant velocity of the Hoeken-type linkage over various fractions $\Delta\beta$ of the crank cycle as a function of the link ratios. [19] The structural error in position (i.e., straightness) ε_S and the structural error in velocity ε_V are defined using notation from Figure 3-29 (p. 121) as:

$$\varepsilon_S = \frac{MAX_{i=1}^{n}\left(C_{y_i}\right) - MIN_{i=1}^{n}\left(C_{y_i}\right)}{\Delta x}$$

$$\varepsilon_V = \frac{MAX_{i=1}^{n}\left(V_{x_i}\right) - MIN_{i=1}^{n}\left(V_{x_i}\right)}{\overline{V}_x} \qquad (3.5)^*$$

The structural errors were computed separately for each of nine crank angle ranges $\Delta\beta$ from 20° to 180°. Table 3-1 shows the link ratios that give the smallest possible structural error in either position or velocity over values of $\Delta\beta$ from 20° to 180°. Note that one cannot attain optimum straightness and minimum velocity error in the same linkage. However, reasonable compromises between the two criteria can be achieved, especially for smaller ranges of crank angle. The errors in both straightness and velocity increase as longer portions of the curve are used (larger $\Delta\beta$). The use of Table 3-1 to design a straight-line linkage will be shown with an example.

* See reference [19] for the derivation of equations 3.5.

TABLE 3-1 Link Ratios for Smallest Attainable Errors in Straightness and Velocity for Various Crank-Angle Ranges of a Hoeken-Type Fourbar Approximate Straight-Line Linkage (19)

Range of Motion			Optimized for Straightness						Optimized for Constant Velocity					
$\Delta\beta$ (deg)	θ_{start} (deg)	% of cycle	Minimum ΔC_y %	ΔV %	$\frac{V_x}{(L_2\omega_2)}$	Link Ratios L_1/L_2	L_3/L_2	$\Delta x/L_2$	Minimum ΔV_x %	ΔC_y %	$\frac{V_x}{(L_2\omega_2)}$	Link Ratios L_1/L_2	L_3/L_2	$\Delta x/L_2$
20	170	5.6%	0.00001%	0.38%	1.436	2.975	3.963	0.601	0.006%	0.137%	1.045	2.075	2.613	0.480
40	160	11.1%	0.00004%	1.53%	1.504	2.950	3.925	1.193	0.038%	0.274%	1.124	2.050	2.575	0.950
60	150	16.7%	0.00027%	3.48%	1.565	2.900	3.850	1.763	0.106%	0.387%	1.178	2.025	2.538	1.411
80	140	22.2%	0.001%	6.27%	1.611	2.825	3.738	2.299	0.340%	0.503%	1.229	1.975	2.463	1.845
100	130	27.8%	0.004%	9.90%	1.646	2.725	3.588	2.790	0.910%	0.640%	1.275	1.900	2.350	2.237
120	120	33.3%	0.010%	14.68%	1.679	2.625	3.438	3.238	1.885%	0.752%	1.319	1.825	2.238	2.600
140	110	38.9%	0.023%	20.48%	1.702	2.500	3.250	3.623	3.327%	0.888%	1.347	1.750	2.125	2.932
160	100	44.4%	0.047%	27.15%	1.717	2.350	3.025	3.933	5.878%	1.067%	1.361	1.675	2.013	3.232
180	90	50.0%	0.096%	35.31%	1.725	2.200	2.800	4.181	9.299%	1.446%	1.374	1.575	1.863	3.456

EXAMPLE 3-12

Designing a Hoeken-Type Straight-Line Linkage.

Problem: A 100-mm-long straight line motion is needed over 1/3 of the total cycle (120° of crank rotation). Determine the dimensions of a Hoeken-type linkage that will

(a) Provide minimum deviation from a straight line. Determine its maximum deviation from constant velocity.

(b) Provide minimum deviation from constant velocity. Determine its maximum deviation from a straight line.

Solution: (see Figure 3-30, p. 123, and Table 3-1)

1 Part (a) requires the most accurate straight line. Enter Table 3-1 at the 6th row which is for a crank angle duration $\Delta\beta$ of the required 120°. The 4th column shows the minimum possible deviation from straight to be 0.01% of the length of the straight line portion used. For a 100-mm length the absolute deviation will then be 0.01 mm (0.0004 in). The 5th column shows that its velocity error will be 14.68% of the average velocity over the 100-mm length. The absolute value of this velocity error of course depends on the speed of the crank.

2 The linkage dimensions for part (a) are found from the ratios in columns 7, 8, and 9. The crank length required to obtain the 100-mm length of straight line Δx is:

from Table 3-1:
$$\frac{\Delta x}{L_2} = 3.238$$
$$L_2 = \frac{\Delta x}{3.238} = \frac{100 \text{ mm}}{3.23} = 30.88 \text{ mm} \tag{a}$$

The other link lengths are then:

from Table 3-1:

$$\frac{L_1}{L_2} = 2.625$$

$$L_1 = 2.625L_2 = 2.625(30.88 \text{ mm}) = 81.07 \text{ mm} \tag{b}$$

from Table 3-1:

$$\frac{L_3}{L_2} = 3.438$$

$$L_3 = 3.438L_2 = 3.438(30.88 \text{ mm}) = 106.18 \text{ mm} \tag{c}$$

The complete linkage is then: $L_1 = 81.07$, $L_2 = 30.88$, $L_3 = L_4 = BP = 106.18$ mm. The nominal velocity V_x of the coupler point at the center of the straight line ($\theta_2 = 180°$) can be found from the factor in the 6th column which must be multiplied by the crank length L_2 and the crank angular velocity ω_2 in radians per second (rad/sec).

3 Part (b) requires the most accurate velocity. Again enter Table 3-1 at the 6th row which is for a crank angle duration $\Delta\beta$ of the required 120°. The 10th column shows the minimum possible deviation from constant velocity to be 1.885% of the average velocity V_x over the length of the straight line portion used. The 11th column shows the deviation from straight to be 0.752% of the length of the straight line portion used. For a 100-mm length the absolute deviation in straightness for this optimum constant velocity linkage will then be 0.75 mm (0.030 in).

The link lengths for this mechanism are found in the same way as was done in step 2 above except that the link ratios 1.825, 2.238, and 2.600 from columns 13, 14, and 15 are used. The result is: $L_1 = 70.19$, $L_2 = 38.46$, $L_3 = L_4 = BP = 86.08$ mm. The nominal velocity V_x of the coupler point at the center of the straight line ($\theta_2 = 180°$) can be found from the factor in the 12th column which must be multiplied by the crank length L_2 and the crank angular velocity ω_2 in rad/sec.

4 The first solution (step 2) gives an extremely accurate straight line over a significant part of the cycle but its 15% deviation in velocity would probably be unacceptable if that factor was considered important. The second solution (step 3) gives less than 2% deviation from constant velocity, which may be viable for a design application. Its 3/4% deviation from straightness, while much greater than the first design, may be acceptable in some situations.

3.9 DWELL MECHANISMS

A common requirement in machine design problems is the need for a dwell in the output motion. A **dwell** is defined as *zero output motionfor some nonzero input motion.* In other words, the motor keeps going, but the output link stops moving. Many production machines perform a series of operations which involve feeding a part or tool into a workspace, and then holding it there (in a dwell) while some task is performed. Then the part must be removed from the workspace, and perhaps held in a second dwell while the rest of the machine "catches up" by indexing or performing some other tasks. Cams and followers (Chapter 8) are often used for these tasks because it is trivially easy to create a dwell with a earn. But, there is always a trade-off in engineering design, and cams have their problems of high cost and wear as described in Section 2.15 (p. 55).

It is also possible to obtain dwells with "pure" linkages of only links and pin joints, which have the advantage over cams of low cost and high reliability. Dwell linkages are more difficult to design than are cams with dwells. Linkages will usually yield only an approximate dwell but will be much cheaper to make and maintain than cams. Thus they may be well worth the effort.

Single-Dwell Linkages

There are two usual approaches to designing single-dwell linkages. Both result in **sixbar mechanisms,** and both require first finding a fourbar with a suitable coupler curve. A **dyad** is then added to provide an output link with the desired dwell characteristic. The first approach to be discussed requires the design or definition of a fourbar with a coupler curve that contains an approximate circle arc portion, which "are" occupies the desired portion of the input link (crank) cycle designated as the dwell. An atlas of coupler curves is invaluable for this part of the task. Symmetrical coupler curves are also well suited to this task, and the information in Figure 3-21 (p. 110) can be used to find them.

✍ EXAMPLE 3-13

Single-Dwell Mechanism with Only Revolute Joints.

Problem: Design a sixbar linkage for 90° rocker motion over 300 crank degrees with dwell for the remaining 60°.

Solution: (see Figure 3-31)

I Search the H&N atlas for a fourbar linkage with a coupler curve having an approximate (pseudo) circle arc portion which occupies 60° of crank motion (12 dashes). The chosen fourbar is shown in Figure 3-3Ia.

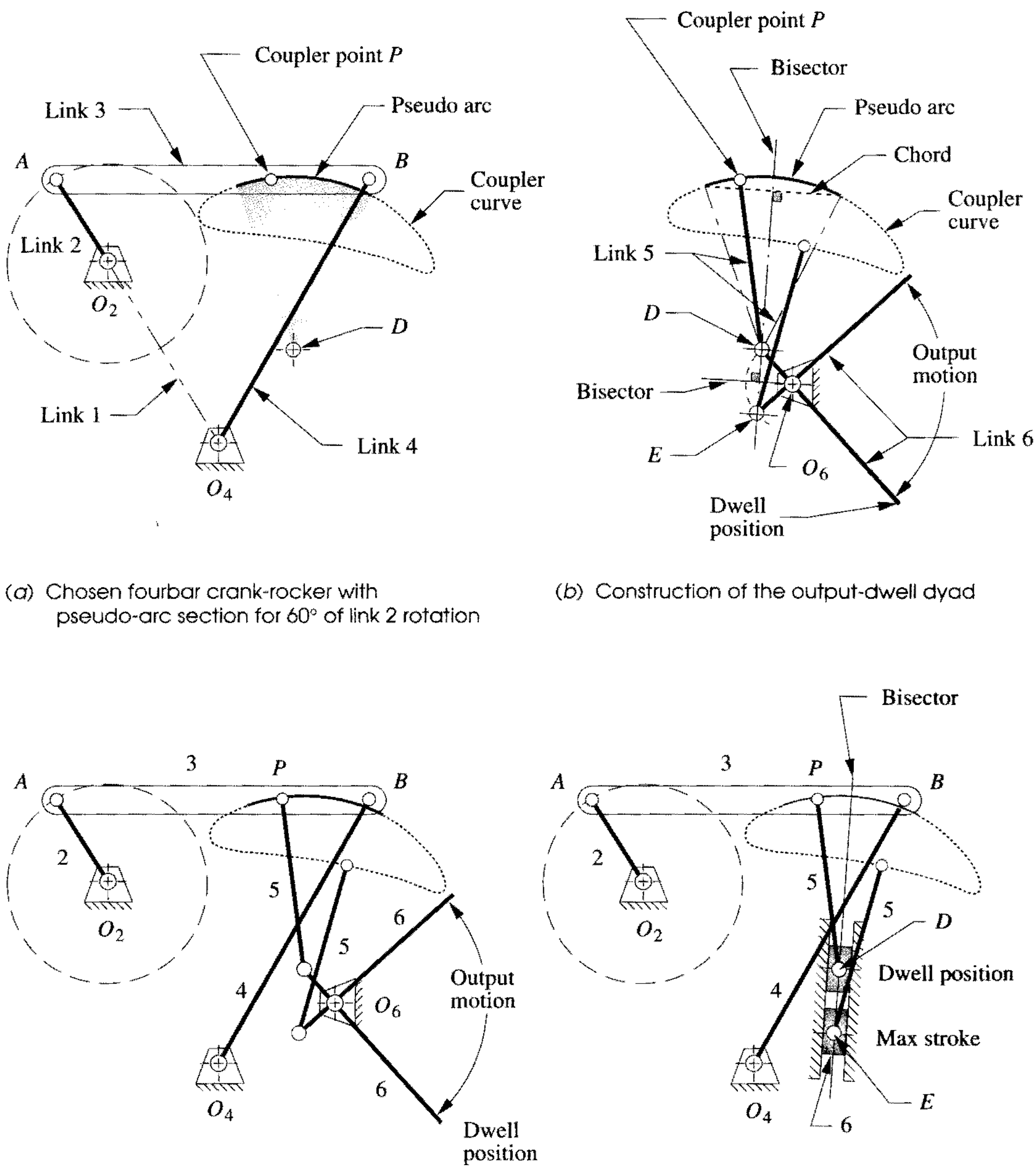
2 Layout this linkage to scale including the coupler curve and find the approximate center of the chosen coupler curve pseudo-arc using graphical geometric techniques. To do so, draw the chord of the arc and construct its perpendicular bisector as shown in Figure 3-31b. The center will lie on the bisector. Label this point D.

3 Set your compass to the approximate radius of the coupler arc. This will be the length of link 5 which is to be attached at the coupler point *P.*

4 Trace the coupler curve with the compass point, while keeping the compass pencil lead on the perpendicular bisector, and find the extreme location along the bisector that the compass lead will reach. Label this point *E.*

5 The line segment *DE* represents the maximum displacement that a link of length *CD,* attached at *P,* will reach along the bisector.

6 Construct a perpendicular bisector of the line segment *DE,* and extend it in a convenient direction.



(a) Chosen fourbar crank-rocker with pseudo-arc section for 60° of link 2 rotation

(b) Construction of the output-dwell dyad

(c) Completed sixbar single-dwell linkage with rocker output option

(d) Completed sixbar single-dwell linkage with slider output option

FIGURE 3-31

Design of a sixbar single-dwell mechanism with rocker output or slider output, using a pseudo-arc coupler curve

7 Locate fixed pivot O_6 on the bisector of DE such that lines O_6D and O_6E subtend the desired output angle, in this example, 90°.

8 Draw link 6 from D (or E) through O_6 and extend to any convenient length. This is the output link which will dwell for the specified portion of the crank cycle.

9 Check the transmission angles.

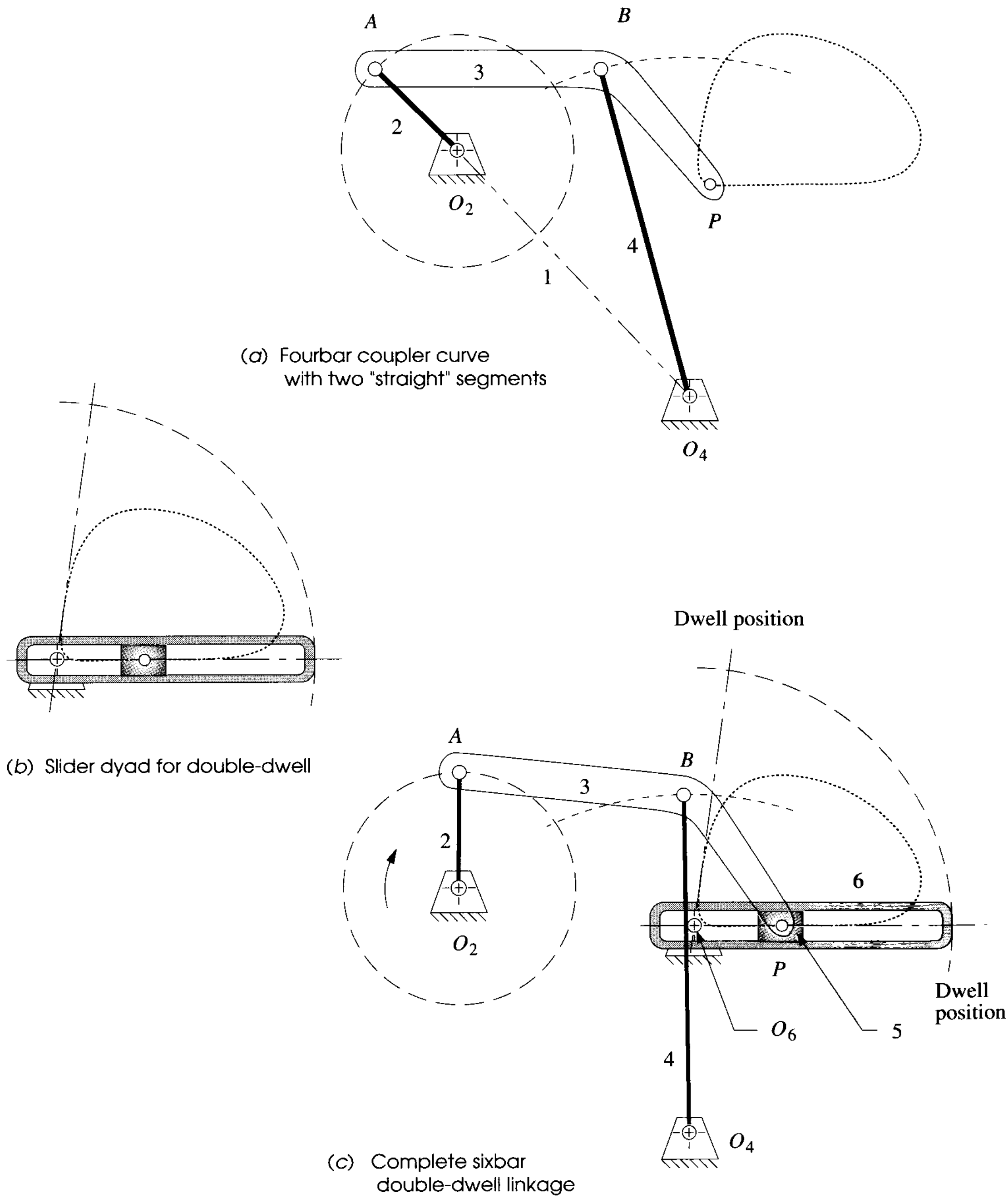
10 Make a cardboard model of the linkage and articulate it to check its function.

This linkage dwells because, during the time that the coupler point P is traversing the pseudo-arc portion of the coupler curve, the other end of link 5, attached to P and the same length as the arc radius, is essentially stationary at its other end, which is the arc center. However the dwell at point D will have some "jitter" or oscillation, due to the fact that D is only an approximate center of the pseudo-arc on the sixth-degree coupler curve. When point P leaves the arc portion, it will smoothly drive link 5 from point D to point E, which will in turn rotate the output link 6 through its arc as shown in Figure 3-31c (p. 127). Note that we can have any angular displacement of link 6 we desire with the same links 2 to 5, as they alone completely define the dwell aspect. Moving pivot O_6 left and right along the bisector of line DE will change the angular displacement of link 6 but not its timing. In fact, a slider block could be substituted for link 6 as shown in Figure 3-31d, and linear translation along line DE with the same timing and dwell at D will result. Input the file F03-31c.6br to program SIXBARand animate to see the linkage of Example 3-13 in motion. The dwell in the motion of link 6 can be clearly seen in the animation, including the jitter due to its approximate nature.

Double-Dwell Linkages

It is also possible, using a fourbar coupler curve, to create a double-dwell output motion. One approach is the same as that used in the single-dwell of Example 3-11. Now a coupler curve is needed which has *two* approximate circle arcs of the same radius but with different centers, both convex or both concave. A link 5 of length equal to the radius of the two arcs will be added such that it and link 6 will remain nearly stationary at the center of each of the arcs, while the coupler point traverses the circular parts of its path. Motion of the output link 6 will occur only when the coupler point is between those arc portions. Higher-order linkages, such as the geared fivebar, can be used to create multiple-dwell outputs by a similar technique since they possess coupler curves with multiple, approximate circle arcs. See the built-in example double-dwell linkage in program SIXBAR for a demonstration of this approach.

A second approach uses a coupler curve with two approximate straight-line segments of appropriate duration. If a pivoted slider block (link 5) is attached to the coupler at this point, and link 6 is allowed to slide in link 5, it only remains to choose a pivot O_6 at the intersection of the straight-line segments extended. The result is shown in Figure 3-32. While block 5 is traversing the "straight-line" segments of the curve, it will not impart any angular motion to link 6. The approximate nature of the fourbar straight line causes some jitter in these dwells also.



(a) Fourbar coupler curve with two "straight" segments

(b) Slider dyad for double-dwell

(c) Complete sixbar double-dwell linkage

FIGURE 3-32

Double-dwell sixbar linkage

EXAMPLE 3-14

Double-Dwell Mechanism.

Problem: Design a sixbar linkage for 80° rocker output motion over 20 crank degrees with dwell for 160°, return motion over 140° and second dwell for 40°.

Solution: (see Figure 3-32, p. 129)

1 Search the H&N atlas for a fourbar linkage with a coupler curve having two approximate straight-line portions. One should occupy 160° of crank motion (32 dashes), and the second 40° of crank motion (8 dashes). This is a wedge-shaped curve as shown in Figure 3-32a.

2 Lay out this linkage to scale including the coupler curve and find the intersection of two tangent lines colinear with the straight segments. Label this point O_6.

3 Design link 6 to lie along these straight tangents, pivoted at O_6. Provide a slot in link 6 to accommodate slider block 5 as shown in Figure 3-32b.

4 Connect slider block 5 to the coupler point P on link 3 with a pin joint. The finished sixbar is shown in Figure 3-32c.

5 Check the transmission angles.

It should be apparent that these linkage dwell mechanisms have some disadvantages. Besides being difficult to synthesize, they give only approximate dwells which have some jitter on them. Also, they tend to be large for the output motions obtained, so do not package well. The acceleration of the output link can also be very high as in Figure 3-32 (p. 129), when block 5 is near pivot O_6. (Note the large angular displacement of link 6 resulting from a small motion of link 5.) Nevertheless they may be of value in situations where a completely stationary dwell is not required, and the low cost and high reliability of a linkage are important factors. Program SIXBAR has both single-dwell and double-dwell example linkages built in.

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3.12 PROBLEMS

*3-1 Define the following examples as path, motion, or function generation cases.

a. A telescope aiming (star tracking) mechanism
b. A backhoe bucket control mechanism
c. A thermostat adjusting mechanism
d. A computer printer head moving mechanism
e. An XY plotter pen control mechanism

3-2 Design a fourbar Grashof crank-rocker for 90° of output rocker motion with no quick-return. (See Example 3-1.) Build a cardboard model and determine the toggle positions and the minimum transmission angle.

*3-3 Design a fourbar mechanism to give the two positions shown in Figure P3-1 of output rocker motion with no quick-return. (See Example 3-2.) Build a cardboard model and determine the toggle positions and the minimum transmission angle.

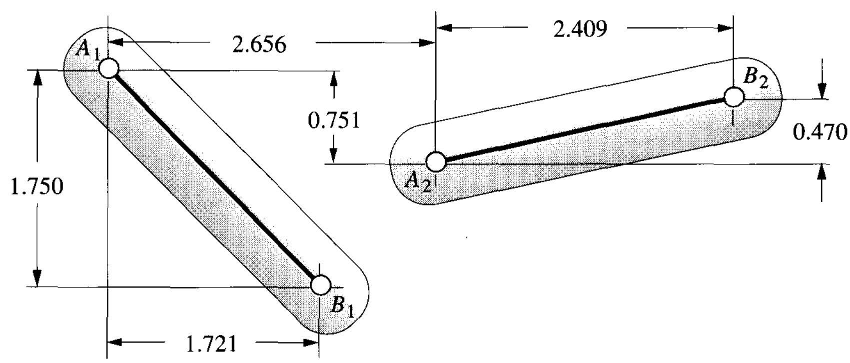


FIGURE P3-1

Problems 3-3 to 3-4

* Answers in Appendix F.

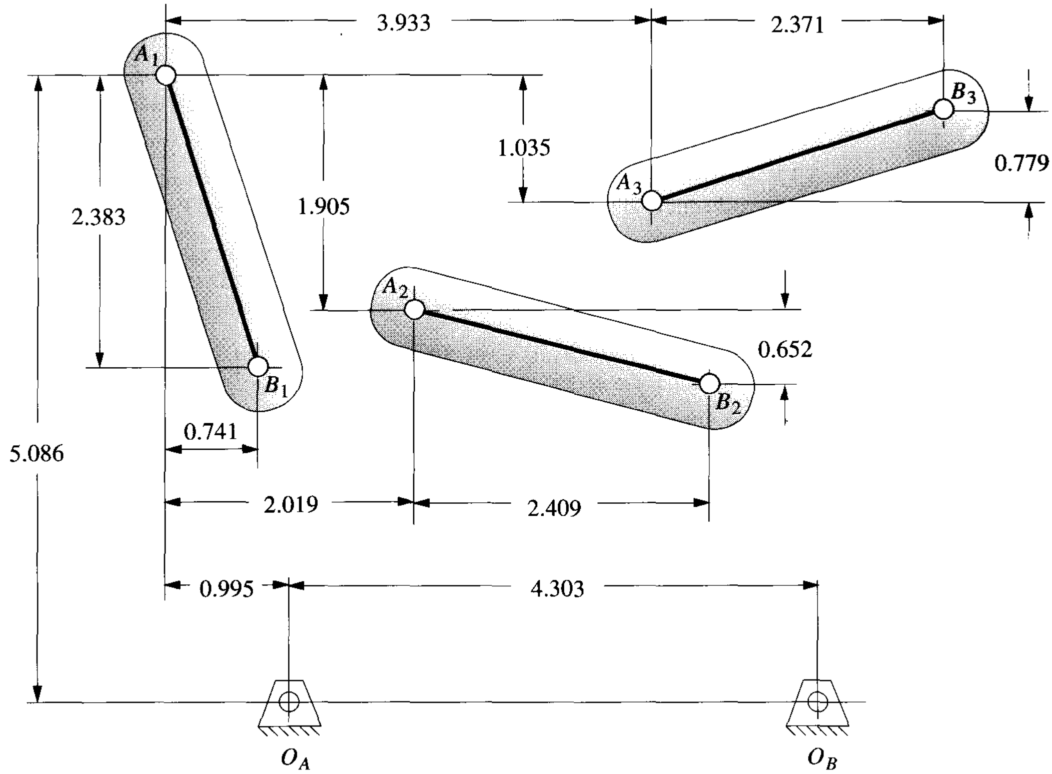


FIGURE P3-2
Problems 3-5 to 3-7

3-4 Design a fourbar mechanism to give the two positions shown in Figure P3-1 of coupler motion. (See Example 3-3.) Build a cardboard model and determine the toggle positions and the minimum transmission angle. Add a driver dyad. (See Example 3-4.)

*3-5 Design a fourbar mechanism to give the three positions of coupler motion with no quick-return shown in Figure P3-2. (See also Example 3-5.) Ignore the points *OA* and *OB* shown. Build a cardboard model and determine the toggle positions and the minimum transmission angle. Add a driver dyad. (See Example 3-4.)

*3-6 Design a fourbar mechanism to give the three positions shown in Figure P3-2 using the fixed pivots *OA* and *OB* shown. Build a cardboard model and determine the toggle positions and the minimum transmission angle. Add a driver dyad.

3-7 Repeat Problem 3-2 with a quick-return time ratio of 1:1.4. (See Example 3-9.)

*3-8 Design a sixbar drag link quick-return linkage for a time ratio of 1:2, and output rocker motion of 60°.

3-9 Design a crank shaper quick-return mechanism for a time ratio of 1 : 3 (Figure 3-14, p. 102).

*3-10 Find the two cognates of the linkage in Figure 3-17 (p. 106). Draw the Cayley and Roberts diagrams. Check your results with program FOURBAR.

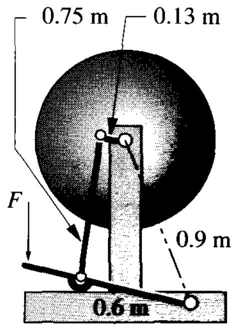


FIGURE P3-3
Problem 3-14. Treadle operated grinding wheel

3-11 Find the three equivalent geared fivebar linkages for the three fourbar cognates in Figure 3-25a (p. 115). Check your results by comparing the coupler curves with programs FOURBAR and FIVEBAR.

3-12 Design a sixbar single-dwell linkage for a dwell of 90° of crank motion, with an output rocker motion of 45°.

3-13 Design a sixbar double-dwell linkage for a dwell of 90° of crank motion, with an output rocker motion of 60°, followed by a second dwell of about 60° of crank motion.

3-14 Figure P3-3 shows a treadle operated grinding wheel driven by a fourbar linkage. Make a cardboard model of the linkage to any convenient scale. Determine its minimum transmission angles. Comment on its operation. Will it work? If so, explain how it does.

3-15 Figure P3-4 shows a non-Grashof fourbar linkage that is driven from link O_2A. All dimensions are in centimeters (cm).

(a) Find the transmission angle at the position shown.
(b) Find the toggle positions in terms of angle AO_2O_4.
(c) Find the maximum and minimum transmission angles over its range of motion.
(d) Draw the coupler curve of point P over its range of motion.

3-16 Draw the Roberts diagram for the linkage in Figure P3-4 and find its two cognates. Are they Grashof or non-Grashof?

3-17 Design a Watt-I sixbar to give parallel motion that follows the coupler path of point P of the linkage in Figure P3-4.

3-18 Add a driver dyad to the solution of Problem 3-17 to drive it over its possible range of motion with no quick return. (The result will be an 8-bar linkage.)

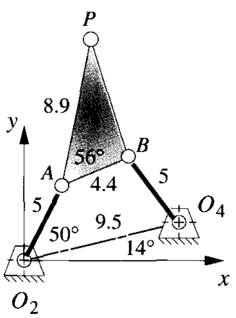


FIGURE P3-4
Problems 3-15 to 3-18

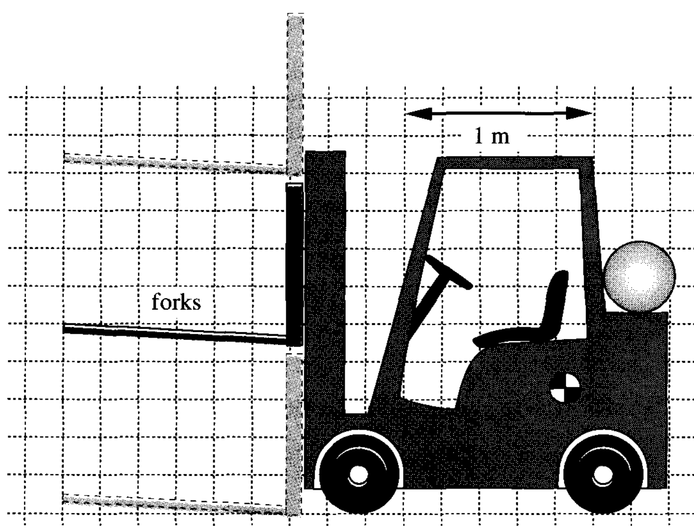


FIGURE P3-5
Problem 3-19

* Answers in Appendix F.

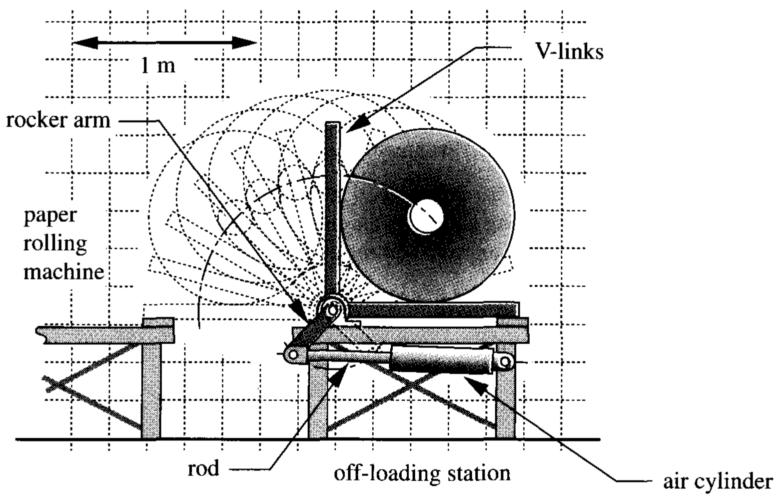


FIGURE P3-6

Problem 3-20

3-19 Design a pin-jointed linkage that will guide the forks of the fork lift truck in Figure P3-5 up and down in an approximate straight line over the range of motion shown. Arrange the fixed pivots so they are close to some part of the existing frame or body of the truck.

3-20 Figure P3-6 shows a "V-link" off-loading mechanism for a paper roll conveyor. Design a pin-jointed linkage to replace the air cylinder driver that will rotate the rocker arm and V-link through the 90° motion shown. Keep the fixed pivots as close to the existing frame as possible. Your fourbar linkage should be Grashof and be in toggle at each extreme position of the rocker arm.

3-21 Figure P3-7 shows a walking-beam transport mechanism that uses a fourbar coupler curve, replicated with a parallelogram linkage for parallel motion. Note the duplicate crank and coupler shown ghosted in the right half of the mechanism—they are redundant

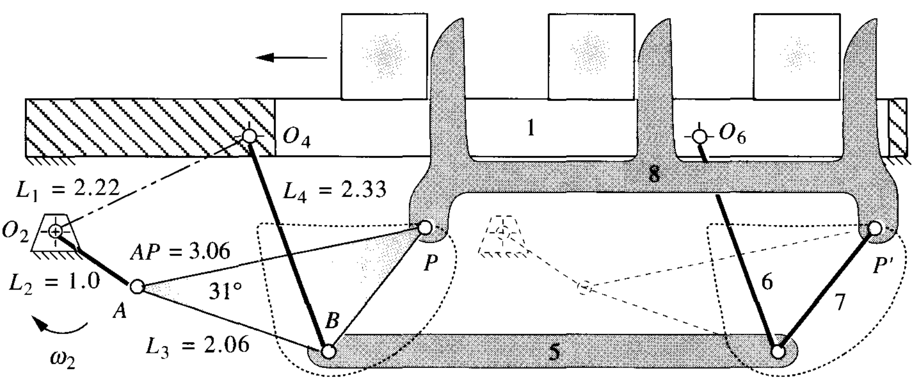


FIGURE P3-7

Problems 3-21 to 3-22. Straight-line walking beam eightbar transport mechanism

* Answers in Appendix F.

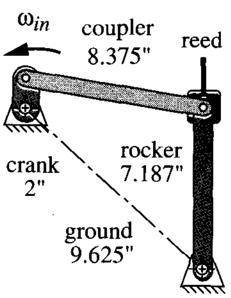


FIGURE P3-8
Problem 3-23.
Loom laybar drive

and have been removed from the duplicate fourbar linkage. Using the same fourbar driving stage (links L_1, L_2, L_3, L_4 with coupler point P), design a Watt-I sixbar linkage that will drive link 8 in the same parallel motion using two fewer links.

*3-22 Find the maximum and minimum transmission angles of the fourbar driving stage (links L_1, L_2, L_3, L_4) in Figure P3-7 (to graphical accuracy).

*3-23 Figure P3-8 shows a fourbar linkage used in a power loom to drive a comblike reed against the thread, "beating it up" into the cloth. Determine its Grashof condition and its minimum and maximum transmission angles to graphical accuracy.

3-24 Draw the Roberts diagram and find the cognates of the linkage in Figure P3-9.

3-25 Find the equivalent geared fivebar mechanism cognate of the linkage in Figure P3-9.

3-26 Use the linkage in Figure P3-9 to design a sixbar double-dwell mechanism that has a rocker output through 45°.

3-27 Use the linkage in Figure P3-9 to design a sixbar double-dwell mechanism that has a slider output stroke of 5 crank units.

3-28 Use two of the cognates in Figure 3-26 (p. 116) to design a Watt-I sixbar parallel motion mechanism that carries a link through the same coupler curve at all points. Comment on its similarities to the original Roberts diagram.

3-29 Find the cognates of the Watt straight-line mechanism in Figure 3-29a (p. 121).

3-30 Find the cognates of the Roberts straight-line mechanism in Figure 3-29b (p. 121).

3-31 Design a Hoeken straight-line linkage to give minimum error in velocity over 22% of the cycle for a 15-cm-long straight-line motion. Specify all linkage parameters.

3-32 Design a Hoeken straight-line linkage to give minimum error in straightness over 39% of the cycle for a 20-cm-long straight-line motion. Specify all linkage parameters.

3-33 Design a linkage that will give a symmetrical "kidney bean" shaped coupler curve as shown in Figure 3-16 (p. 104 and 105). Use the data in Figure 3-21 (p. 110) to determine the required link ratios and generate the coupler curve with program FOURBAR.

3-34 Repeat Problem 3-33 for a "double straight" coupler curve.

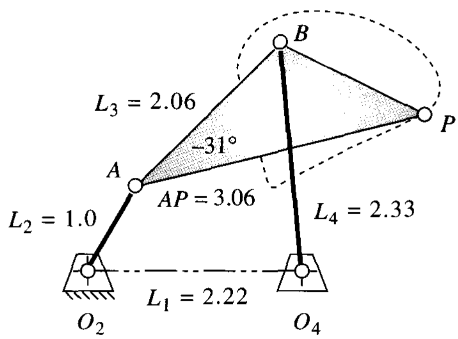


FIGURE P3-9
Problems 3-24 to 3-27

* Answers in Appendix F.

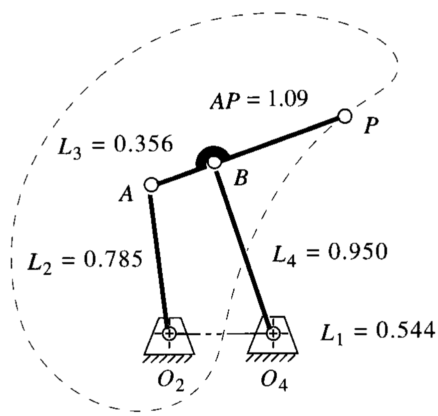


FIGURE P3-10

Problems 3-36 to 3-38

3-35 Repeat problem 3-33 for a "scimitar" coupler curve with two distinct cusps. Show that there are (or are not) true cusps on the curve by using program FOURBAR. (Hint: Think about the definition of a cusp and how you can use the program's data to show it.)

*3-36 Find the Grashof condition, inversion, any limit positions, and the extreme values of the transmission angle (to graphical accuracy) of the linkage in Figure P3-10.

3-37 Draw the Roberts diagram and find the cognates of the linkage in Figure P3-10.

3-38 Find the three geared fivebar cognates of the linkage in Figure P3-10.

3-39 Find the Grashof condition, any limit positions, and the extreme values of the transmission angle (to graphical accuracy) of the linkage in Figure P3-11.

3-40 Draw the Roberts diagram and find the cognates of the linkage in Figure P3-11.

3-41 Find the three geared fivebar cognates of the linkage in Figure P3-11.

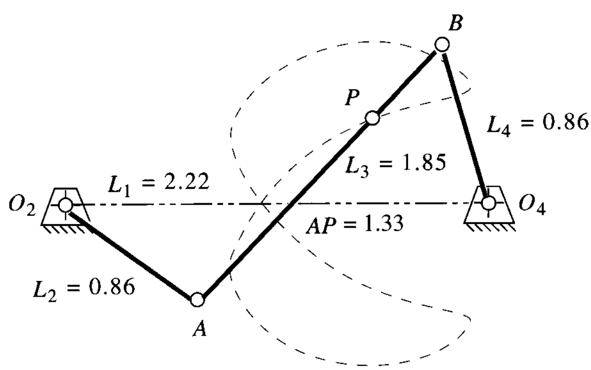


FIGURE P3-11

Problems 3-39 to 3-41

* Answers in Appendix F.

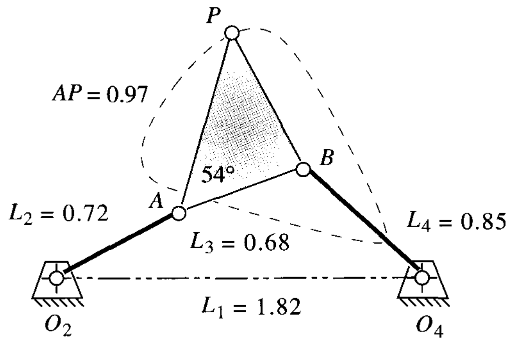


FIGURE P3-12

Problems 3-42 to 3-44

*3-42 Find the Grashof condition, any limit positions, and the extreme values of the transmission angle (to graphical accuracy) of the linkage in Figure P3-12.

3-43 Draw the Roberts diagram and find the cognates of the linkage in Figure P3-12.

3-44 Find the three geared fivebar cognates of the linkage in Figure P3-12.

3-45 Prove that the relationships between the angular velocities of various links in the Roberts diagram as shown in Figure 3-25 (p. 115) are true.

3-46 Design a fourbar linkage to move the object in Figure P3-13 from position 1 to 2 using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions shown making it a sixbar. All fixed pivots should be on the base.

3-47 Design a fourbar linkage to move the object in Figure P3-13 from position 2 to 3 using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions shown making it a sixbar. All fixed pivots should be on the base.

3-48 Design a fourbar linkage to move the object in Figure P3-13 through the three positions shown using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions shown making it a sixbar. All fixed pivots should be on the base.

3-49 Design a fourbar linkage to move the object in Figure P3-14 from position 1 to 2 using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions shown making it a sixbar. All fixed pivots should be on the base.

3-50 Design a fourbar linkage to move the object in Figure P3-14 from position 2 to 3 using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions shown making it a sixbar. All fixed pivots should be on the base.

3-51 Design a fourbar linkage to move the object in Figure P3-14 through the three positions shown using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions shown making it a sixbar. All fixed pivots should be on the base.

3-52 Design a fourbar linkage to move the object in Figure P3-15 from position 1 to 2 using points *A* and *B* for attachment. Add a driver dyad to limit its motion to the range of positions shown making it a sixbar. All fixed pivots should be on the base.

* Answers in Appendix F.

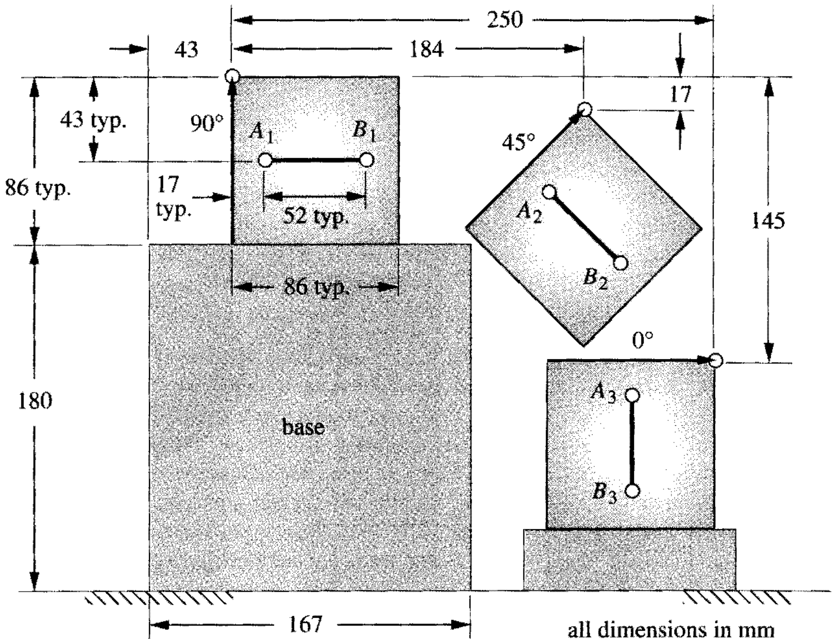


FIGURE P3-13

Problems 3-46 to 3-48

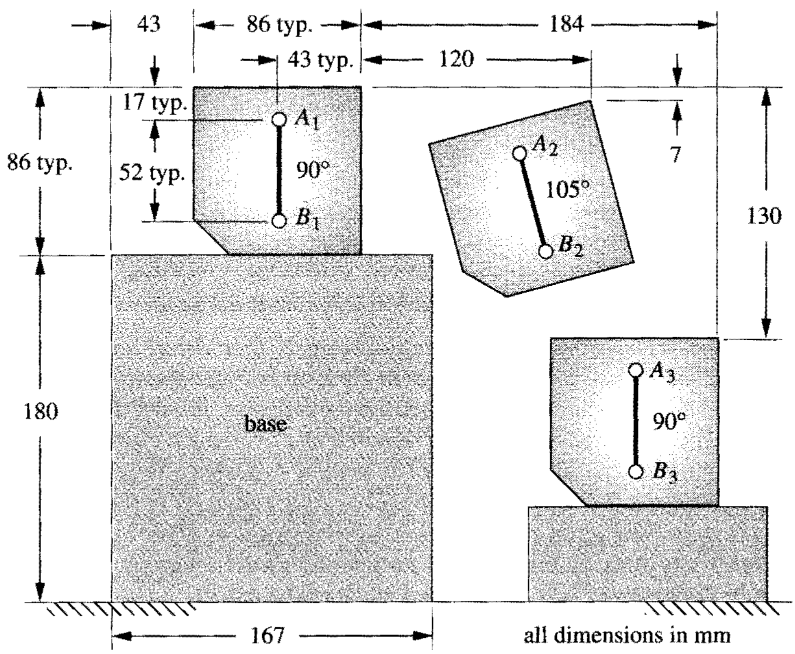


FIGURE P3-14

Problems 3-49 to 3-51

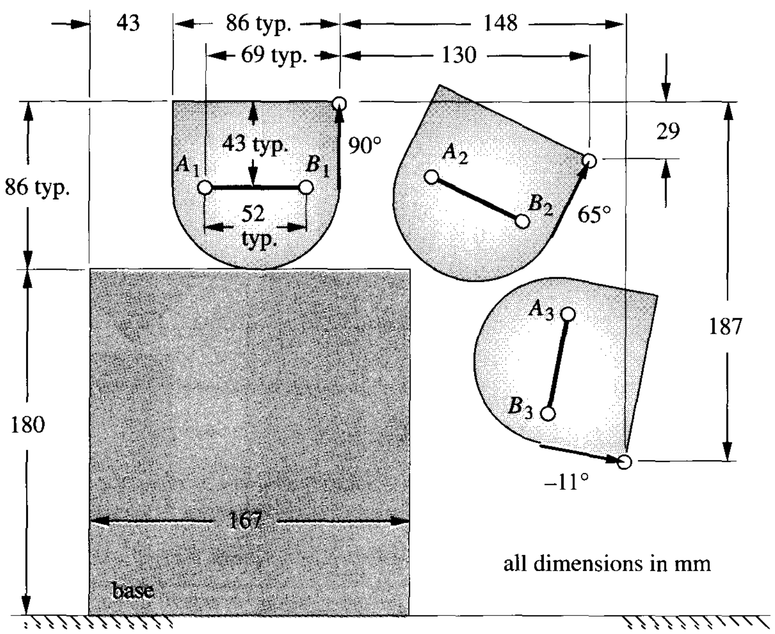


FIGURE P3-15
Problems 3-52 to 3-54

3-53 Design a fourbar linkage to move the object in Figure P3-15 from position 2 to 3 using points A and B for attachment. Add a driver dyad to limit its motion to the range of positions shown making it a sixbar. All fixed pivots should be on the base.

3-54 Design a fourbar linkage to move the object in Figure P3-15 through the three positions shown using points A and B for attachment. Add a driver dyad to limit its motion to the range of positions shown making it a sixbar. All fixed pivots should be on the base.

3.13 PROJECTS

These larger-scale project statements deliberately lack detail and structure and are loosely defined. Thus, they are similar to the kind of "identification of need" problem statement commonly encountered in engineering practice. It is left to the student to structure the problem through background research and to create a clear goal statement and set of performance specifications before attempting to design a solution. This design process is spelled out in Chapter 1 and should be followed in all of these examples. These projects can be done as an exercise in mechanism synthesis alone or can be revisited and thoroughly analyzed by the methods presented in later chapters as well. All results should be documented in a professional engineering report.

P3-1 The tennis coach needs a better tennis ball server for practice. This device must fire a sequence of standard tennis balls from one side of a standard tennis court over the net such that they land and bounce within each of the three court areas defined by the court's white lines. The order and frequency of a ball's landing in any one of the three court areas must be random. The device should operate automatically and unattended except for the refill of balls. It should be capable of firing 50 balls between reloads. The timing of ball releases should vary. For simplicity, a motor driven pin-jointed linkage design is preferred.

P3-2 A quadriplegic patient has lost all motion except that of her head. She can only move a small "mouth stick" to effect a switch closure. She was an avid reader before her injury and would like again to be able to read standard hardcover books without the need of a person to turn pages for her. Thus, a reliable, simple, and inexpensive automatic page tumer is needed. The book may be placed in the device by an assistant. It should accommodate as wide a range of book sizes as possible. Book damage is to be avoided and safety of the user is paramount.

P3-3 Grandma's off her rocker again! Junior's run down to the Bingo parlor to fetch her, but we've got to do something about her rocking chair before she gets back. She's been complaining that her arthritis makes it too painful to push the rocker. So, for her l00th birthday in 2 weeks, we're going to surprise her with a new, automated, motorized rocking chair. The only constraints placed on the problem are that the device must be safe and must provide interesting and pleasant motions, similar to those of her present *Boston* rocker, to all parts of the occupant's body. Since simplicity is the mark of good design, a linkage solution with only full pin joints is preferred.

P3-4 The local amusement park's business is suffering as a result of the proliferation of computer game parlors. They need a new and more exciting ride which will attract new customers. The only constraints are that it must be safe, provide excitement, and not subject the occupants to excessive accelerations or velocities. Also it must be as compact as possible, since space is limited. Continuous rotary input and full pin joints are preferred.

P3-5 The student section of ASME is sponsoring a spring fling on campus. They need a mechanism for their "Dunk the Professor" booth which will carry the unfortunate (untenured) volunteer into and out of the water tub. The contestants will provide the inputs to a multiple-DOF mechanism. If they know their kinematics, they can provide a combination of inputs which will dunk the victim.

P3-6 The National House of Flapjacks wants to automate their flapjack production. They need a mechanism which will automatically flip the flapjacks "on the fly" as they travel through the griddle on a continuously moving conveyor. This mechanism must track the constant velocity of the conveyor, pick up a pancake, flip it over and place it back onto the conveyor.

P3-7 Many varieties and shapes of computer video monitors now exist. Their long-term use leads to eyestrain and body fatigue. There is a need for an adjustable stand which will hold the video monitor and the separate keyboard at any position the user deems comfortable. The computer's central processor unit (CPU) can be remotely located. This device should be freestanding to allow use with a comfortable chair, couch, or lounge of the user's choice. It should not require the user to assume the conventional "seated at a desk" posture to use the computer. It must be stable in all positions and safely support the equipment's weight.

P3-8 Most small boat trailers must be submerged in the water to launch or retrieve the boat. This greatly reduces the life of the trailer, especially in salt water. A need exists for a trailer that will remain on dry land while it launches or retrieves the boat. No part of the trailer should get wet. User safety is of greatest concern, as is protection of the boat from damage.

P3-9 The "Save the Skeet" foundation has requested a more humane skeet launcher be designed. While they have not yet succeeded in passing legislation to prevent the wholesale slaughter of these little devils, they are concerned about the inhumane aspects of the large accelerations imparted to the skeet as it is launched into the sky for the sportsman to shoot it down. The need is for a skeet launcher that will smoothly accelerate the clay pigeon onto its desired trajectory.

P3-10 The coin operated "kid bouncer" machines found outside supermarkets typically provide a very unimaginative rocking motion to the occupant. There is a need for a superior "bouncer" which will give more interesting motions while remaining safe for small children.

P3-11 Horseback riding is a very expensive hobby or sport. There is a need for a horseback riding simulator to train prospective riders sans the expensive horse. This device should provide similar motions to the occupant as she would feel in the saddle under various gaits such as a walk, canter, or gallop. A more advanced version might contain jumping motions as well. User safety is most important.

P3-12 The nation is on a fitness craze. Many exercise machines have been devised. There is still room for improvement to these devices. They are typically designed for the young, strong athlete. There is also a need for an ergonomically optimum exercise machine for the older person who needs gentler exercise.

P3-13 A paraplegic patient needs a device to get himself from his wheelchair into the Jacuzzi with no assistance. He has good upper body and arm strength. Safety is paramount.

P3-14 The army has requested a mechanical walking device to test army boots for durability. It should mimic a person's walking motion and provide forces similar to an average soldier's foot.

P3-15 NASA wants a zero-G machine for astronaut training. It must carry one person and provide a negative I-G acceleration for as long as possible.

P3-16 The Amusement Machine Co. Inc. wants a portable "whip" ride which will give two or four passengers a thrilling but safe ride, and which can be trailed behind a pickup truck from one location to another.

P3-17 The Air Force has requested a pilot training simulator which will give potential pilots exposure to G forces similar to those they will experience in dogfight maneuvers.

P3-18 Cheers needs a better "mechanical bull" simulator for their "yuppie" bar in Boston. It must give a thrilling "bucking bronco" ride but be safe.

P3-19 Despite the improvements in handicap access, many curbs block wheelchairs from public places. Design an attachment for a conventional wheelchair which will allow it to get up over a curb.

P3-20 A carpenter needs a dumping attachment to fit in her pickup truck so she can dump building materials. She can't afford to buy a dump truck.

P3-21 The carpenter in Project P3-20 wants an inexpensive lift gate designed to fit her full-sized pickup truck, in order to lift and lower heavy cargo to the truck bed.

P3-22 The carpenter in Project P3-20 is very demanding (and lazy). She also wants a device to lift Sheet rock into place on ceilings or walls to hold it while she nails it on.

P3-23 Click and Clack, the tappet brothers, need a better transmission jack for their Good News Garage. This device should position a transmission under a car (on a lift) and allow it to be maneuvered into place safely and quickly.

P3-24 A paraplegic who was an avid golfer before his injury, wants a mechanism to allow him to stand up in his wheelchair in order to once again play golf. It must not interfere with normal wheelchair use, though it could be removed from the chair when he is not golfing.

P3-25 A wheelchair lift is needed to raise the wheelchair and person 3 ft from the garage floor to the level of the first floor of the house. Safety and reliability are of major concern, as is cost.

P3-26 A paraplegic needs a mechanism that can be installed on a full-size 3-door pickup truck that will lift the wheelchair into the area behind the driver's seat. This person has excellent upper body strength and, with the aid of specially installed handles on the truck, can get into the cab from the chair. The truck can be modified as necessary to accommodate this task. For example, attachment points can be added to its structure and the back seat of the truck can be removed if necessary.

P3-27 There is demand for a better *Baby Transport Device.* Many such devices are on the market. Some are called carriages, some strollers. Some are convertible to multiple uses. Our marketing survey data so far seems to indicate that the customers want portability (i.e., foldability), light weight, one-handed operation, and large wheels. Some of these features are present in existing devices. We need a better design that more completely meets the needs of the customer. The device must be stable, effective, and safe for the baby and the operator. Full joints are preferred to half joints and simplicity is the mark of good design. A linkage solution with manual input is desired.

P3-28 There is a need for a *dining-table leaf insertion device.* The device must be simple to use, preferably using the action of opening the table-halves as the actuating motion. That is, as you pull the table open, the stored leaf should be carried by the mechanism of your design into its proper place in order to extend the dining surface.

P3-29 A boat owner has requested that we design her a lift mechanism to automatically move a lOOO-lb, 15-ft boat from a cradle on land to the water. A seawall protects the owner's yard, and the boat cradle sits above the seawall. The tidal variation is 4 ft. Your mechanism will be attached to land and move the boat from its stored position on the cradle to the water and return it to the cradle. The device must be safe and easy to use and not overly expensive.

P3-30 The landfills are full! We're about to be up to our ears in trash! The world needs a better trash compactor. It should be simple, inexpensive, quiet, compact, and safe. It can either be manually powered or motorized, but manual operation is preferred to keep the cost down. The device must be stable, effective, and safe for the operator.

P3-31 A small contractor needs a mini-dumpster attachment for his pickup truck. He has made several trash containers which are 4 ft x 4 ft x 3.5 ft high. The empty container weighs ISO lb. He needs a mechanism which he can attach to his fleet of standard, full-size pickup trucks (Chevrolet, Ford, or Dodge). This mechanism should be able to pick up the full trash container from the ground, lift it over the closed tailgate of the truck, dump its contents into the truck bed, and then return it empty to the ground. He would like not to tip his truck over in the process. The mechanism should store permanently on the truck in such a manner as to allow the normal use of the pickup truck at all other times. You may specify any means of attachment of your mechanism to the container and to the truck.

Chapter 4

POSITION ANALYSIS

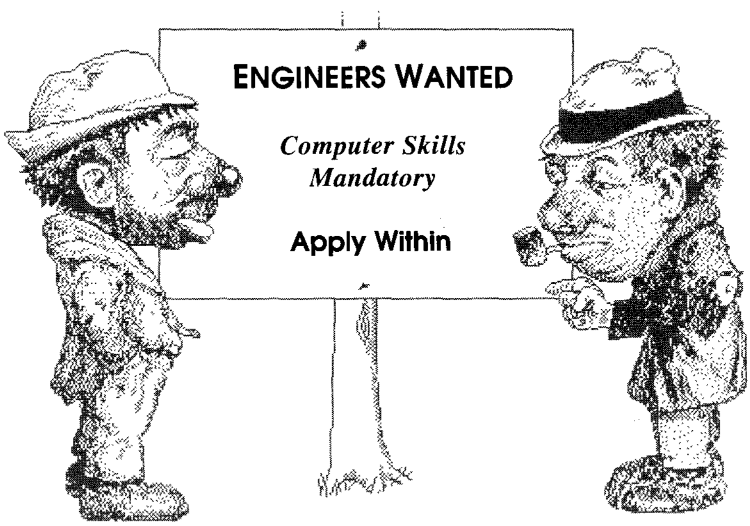
Theory is the distilled essence of practice
RANKINE

4.0 INTRODUCTION

Once a tentative mechanism design has been synthesized, it must then be analyzed. A principal goal of kinematic analysis is to determine the accelerations of all the moving parts in the assembly. Dynamic forces are proportional to acceleration, from Newton's second law. We need to know the dynamic forces in order to calculate the stresses in the components. The design engineer must ensure that the proposed mechanism or machine will not fail under its operating conditions. Thus the stresses in the materials must be kept well below allowable levels. To calculate the stresses, we need to know the static and dynamic forces on the parts. To calculate the dynamic forces, we need to know the accelerations. In order to calculate the accelerations, we must first find the positions of all the links or elements in the mechanism for each increment of input motion, and then differentiate the position equations versus time to find velocities, and then differentiate again to obtain the expressions for acceleration. For example, in a simple Grashof fourbar linkage, we would probably want to calculate the positions, velocities, and accelerations of the output links (coupler and rocker) for perhaps every two degrees (180 positions) of input crank position for one revolution of the crank.

This can be done by any of several methods. We could use a graphical approach to determine the position, velocity, and acceleration ofthe output links for all 180 positions of interest, or we could derive the general equations of motion for any position, differentiate for velocity and acceleration, and then solve these analytical expressions for our 180 (or more) crank locations. A computer will make this latter task much more palatable. If we choose to use the graphical approach to analysis, we will have to do an independent graphical solution for each of the positions of interest. None of the information obtained graphically for the first position will be applicable to the second position or to any others. In contrast, once the analytical solution is derived for a particular mechanism, it can be quickly solved (with a computer) for all positions. If you want information for more than 180 positions, it only means you will have to wait longer for the computer to generate those data. The derived equations are the same. So, have another cup of coffee while the computer crunches the numbers! In this chapter, we will present and derive analytical solutions to the position analysis problem for various planar mechanisms. We will also discuss graphical solutions which are useful for checking your analytical results. In Chapters 6 and 7 we will do the same for velocity and acceleration analysis of planar mechanisms.

It is interesting to note that **graphical position** analysis of linkages is a truly trivial exercise, while the algebraic approach to position analysis is much more complicated. If you can draw the linkage to scale, you have then solved the position analysis problem graphically. It only remains to measure the link angles on the scale drawing to protractor accuracy. But, the converse is true for velocity and especially for acceleration analysis. Analytical solutions for these are less complicated to derive than is the analytical position solution. However, graphical velocity and acceleration analysis becomes quite complex and difficult. Moreover, the graphical vector diagrams must be redone *de novo* (meaning literally *from new)* for each of the linkage positions of interest. This is a very tedious exercise and was the only practical method available in the days *B.c. (Before Computer),* not so long ago. The proliferation of inexpensive microcomputers in recent years has truly revolutionized the practice of engineering. As a graduate engineer, you will never be far from a computer of sufficient power to solve this type of problem and may even have one in your pocket. Thus, in this text we will emphasize analytical solutions which are easily solved with a microcomputer. The computer programs provided with this text use the same analytical techniques as derived in the text.



Geez Joe, - now I wish I took that programming course!

4.1 COORDINATE SYSTEMS

Coordinate systems and reference frames exist for the pleasure and convenience of the engineer who defines them. In the next chapters we will freely adorn our systems with multiple coordinate systems as we see fit, to aid in understanding and solving the problem. We will denote one of these as the *global* or *absolute* coordinate system, and the others will be *local* coordinate systems within the global framework. The global system is often taken to be attached to Mother Earth, though it could as well be attached to another ground plane such as the frame of an automobile. If our goal is to analyze the motion of a windshield wiper blade, we may not care to include the gross motion of the automobile in the analysis. In that case a global coordinate system attached to the car would be useful, and we could consider it to be an **absolute** coordinate system. Even if we use the earth as an absolute reference frame, we must realize that it is not stationary either, and as such is not very useful as a reference frame for a space probe. Though we will speak of absolute positions, velocities, and accelerations, keep in mind that ultimately, until we discover some stationary point in the universe, all motions are really relative. The term **inertial reference frame** is used to denote *a system which itself has no acceleration.* All angles in this text will be measured according to the *right-hand rule.* That is, **counterclockwise** angles, angular velocities, and angular accelerations *are positive in sign.*

Local coordinate systems are typically attached to a link at some point of interest. This might be a pin joint, a center of gravity, or a line of centers of a link. These local coordinate systems may be either rotating or nonrotating as we desire. If we want to measure the angle of a link as it rotates in the global system, we probably will want to attach a nonrotating coordinate system to some point on the link (say a pin joint). This nonrotating system will move with its origin on the link but remains always parallel to the global system. If we want to measure some parameters within a link, independent of its rotation, then we will want to construct a rotating coordinate system along some line on the link. This system will both move and rotate with the link in the global system. Most often we will need to have both types of local coordinate systems on our moving links to do a complete analysis. Obviously we must define the positions and angles of these moving, local coordinate systems in the global system at all positions of interest.

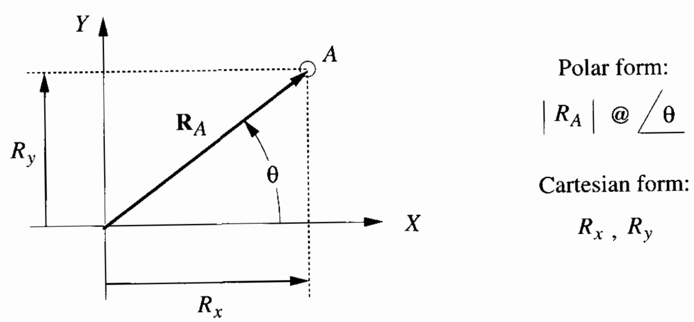


FIGURE 4-1
A position vector in the plane

4.2 POSITION AND DISPLACEMENT

Position

The **position** of a point in the plane can be defined by the use of a **position vector** as shown in Figure 4-1. The choice of **reference axes** is arbitrary and is selected to suit the observer. A two-dimensional vector has two attributes, which can be expressed in either *polar* or *cartesian* coordinates. The **polar form** provides the magnitude and the angle of the vector. The **cartesian form** provides the X and Y components of the vector. Each form is directly convertible into the other by:*

the Pythagorean theorem:

$$R_A = \sqrt{R_x^2 + R_y^2}$$

and trigonometry: (4.0)

$$\theta = \arctan\left(\frac{R_y}{R_x}\right)$$

Displacement

Displacement of a point is the change in its position and can be defined as *the straight-line distance between the initial and final position of a point which has moved in the reference frame.* Note that displacement is not necessarily the same as the path length which the point may have traveled to get from its initial to final position. Figure 4-2a shows a point in two positions, A and B. The curved line depicts the path along which the point traveled. The position vector R_{BA} defines the displacement of the point B with respect to pointA. Figure 4-2b defines this situation more rigorously and with respect to a reference frame XY. The notation R will be used to denote a position vector. The vectors R_A and R_B define, respectively, the absolute positions of points A and B with respect to this *global XY* reference frame. The vector R_{BA} denotes the difference in position, or the *displacement,* between A and B. This can be expressed as the *position difference equation:*

$$R_{BA} = R_B - R_A \tag{4.1a}$$

This expression is read: *The position of B with respect to A is equal to the (absolute) position of B minus the (absolute) position of A,* where *absolute* means with respect to the origin of the *global* reference frame. This expression could also be written as:

$$R_{BA} = R_{BO} - R_{AO} \tag{4.1b}$$

with the second subscript 0 denoting the origin of the XY reference frame. When a position vector is rooted at the origin of the reference frame, it is customary to omit the second subscript. It is understood, in its absence, to be the origin. Also, a vector referred to the origin, such as R_A, is often called an absolute vector. This means that it is taken with respect to a reference frame which is assumed to be stationary, e.g. *the ground.* It is important to realize, however, that the ground is usually also in motion in some larger frame ofreference. Figure 4-2c shows a graphical solution to equation 4.1.

---

* Note that a two-argument arctangent function must be used to obtain angles in all four quadrants. The single argument arctangent function found in most calculators and computer programming languages returns angle values in only the first and fourth quadrants. You can calculate your own two-argument arctangent function very easily by testing the sign of the x component of the arguments and. if x is minus, adding π radians or 180° to the result obtained from the available single-argument arctangent function.

For example (in Fortran):

```
FUNCTION Atan2( x, y )
IF x <> 0 THEN Q = y / x
Temp = ATAN(Q)
IF x < 0 THEN
  Atan2 = Temp + 3.14159
ELSE
  Atan2 = Temp
END IF
RETURN
END
```

The above code assumes that the language used has a built-in single-argument arctangent function called ATAN(x) which returns an angle between $\pm \pi/2$ radians when given a signed argument representing the value of the tangent of that angle.

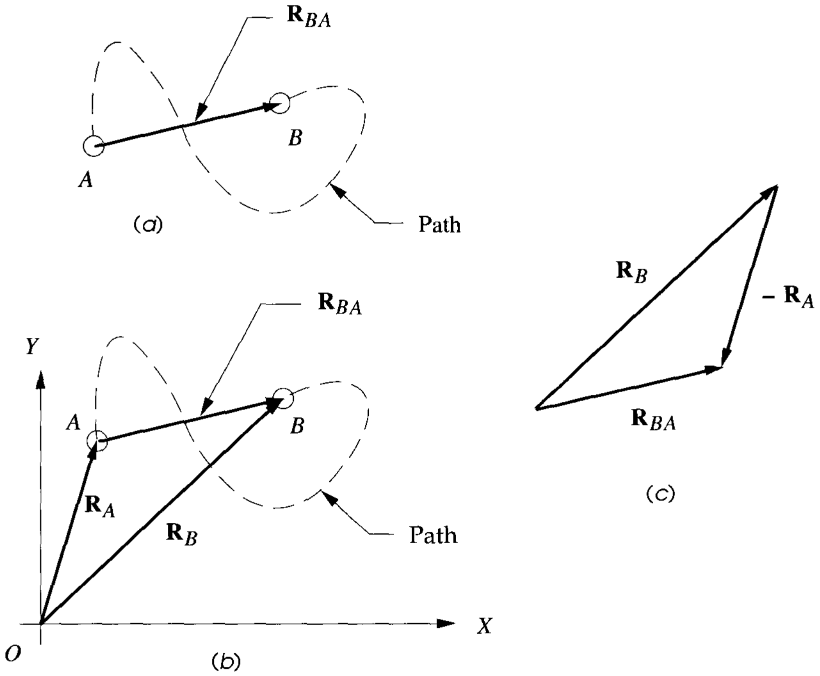


FIGURE 4-2
Position difference and relative position

In our example of Figure 4-2, we have tacitly assumed so far that this point, which is first located at A and later at B, is, in fact, the same particle, moving within the reference frame. It could be, for example, one automobile moving along the road from A to B. With that assumption, it is conventional to refer to the vector R_{BA} as a position difference. There is, however, another situation which leads to the same diagram and equation but needs a different name. Assume now that points A and B in Figure 4-2b represent not the same particle but two independent particles moving in the same reference frame, as perhaps two automobiles traveling on the same road. The vector equations 4.1 and the diagram in Figure 4-2b still are valid, but we now refer to R_{BA} as a relative position, or apparent position. We will use the relative position term here. A more formal way to distinguish between these two cases is as follows:

CASE 1: *One body in two successive positions* => *position difference*

CASE 2: *Two bodies simultaneously in separate positions* => *relative position*

This may seem a rather fine point to distinguish, but the distinction will prove useful, and the reasons for it more clear, when we analyze velocities and accelerations, especially when we encounter (CASE 2 type) situations in which the two bodies occupy the same position at the same time but have different motions.

4.3 TRANSLATION, ROTATION, AND COMPLEX MOTION

So far we have been dealing with a particle, or point, in plane motion. It is more interesting to consider the motion of a **rigid body**, or link. Figure 4-3a shows a link *AB* denoted by a position vector $\mathbf{R}_{BA}$. An axis system has been set up at the root of the vector, at point *A*, for convenience.

Translation

Figure 4-3b shows link *AB* moved to a new position *A'B'* by translation through the displacement *AA'* or *BB'* which are equal, i.e., $\mathbf{R}_{A'A} = \mathbf{R}_{B'B}$.

A definition of translation is:

All points on the body have the same displacement.

As a result the link retains its angular orientation. Note that the translation need not be along a straight path. The curved lines from *A* to *A'* and *B* to *B'* are the **curvilinear translation** path of the link. There is no rotation of the link if these paths are parallel. If the path happens to be straight, then it will be the special case of **rectilinear translation**, and the path and the displacement will be the same.

Rotation

Figure 4-3c shows the same link *AB* moved from its original position at the origin by rotation through an angle. Point *A* remains at the origin, but *B* moves through the position difference vector $\mathbf{R}_{B'B} = \mathbf{R}_{B'A} - \mathbf{R}_{BA}$.

A definition of rotation is:

Different points in the body undergo different displacements and thus there is a displacement difference between any two points chosen.

The link now changes its angular orientation in the reference frame, and all points have different displacements.

Complex Motion

The general case of **complex motion** is the sum of the translation and rotation components. Figure 4-3d shows the same link moved through both the translation and the rotation applied above. Note that the order in which these two components are added is immaterial. The resulting complex displacement will be the same whether you first rotate and then translate or vice versa. This is because the two factors are independent. The total complex displacement of point *B* is defined by the following expression:

Total displacement = translation component + rotation component

$$\mathbf{R}_{B''B} = \mathbf{R}_{B'B} + \mathbf{R}_{B''B'} \qquad (4.1c)$$

The new absolute position of point *B* referred to the origin at *A* is:

$$\mathbf{R}_{B''A} = \mathbf{R}_{A'A} + \mathbf{R}_{B''A'} \qquad (4.1d)$$

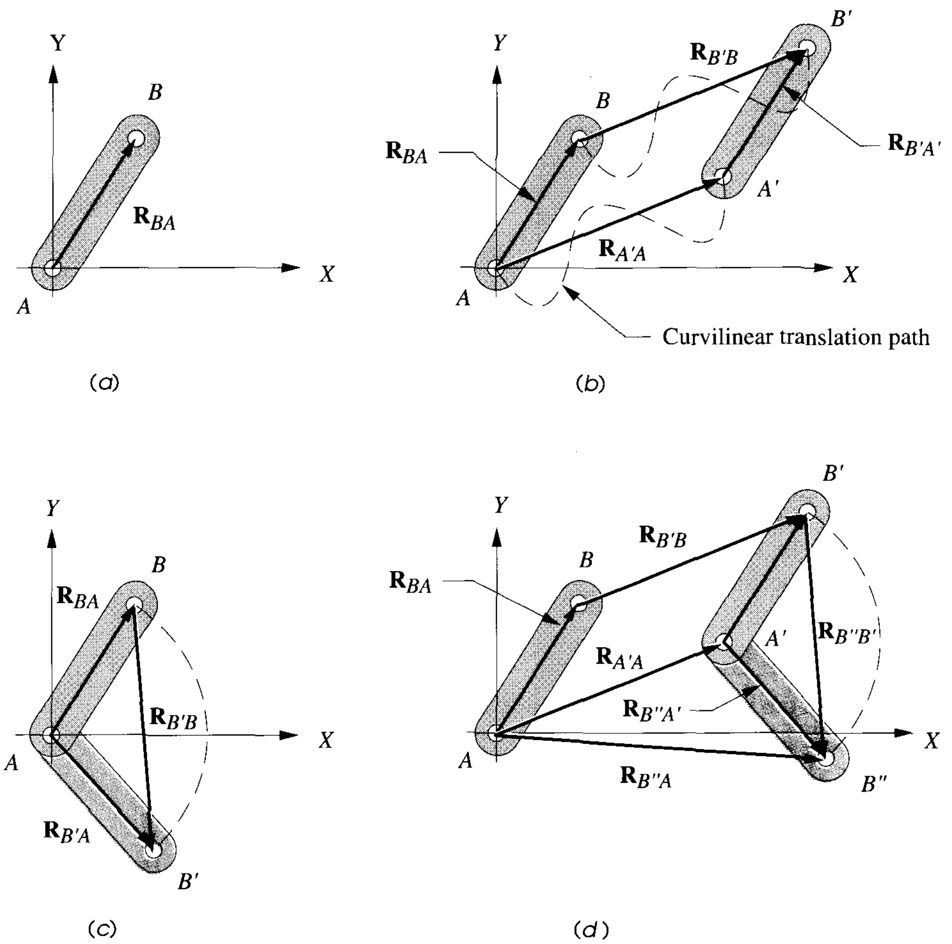


FIGURE 4-3

Translation, rotation, and complex motion

Note that the above two formulas are merely applications of the position difference equation 4.1a. See also Section 2.2 (p. 23) for definitions and discussion of *rotation, translation,* and *complex motion.* These motion states can be expressed as the following theorems.

Theorems

Euler's theorem:

The general displacement of a rigid body with one point fixed is a rotation about some axis.

This applies to pure rotation as defined above and in Section 2.2 (p. 23). Chasles (1793-1880) provided a corollary to Euler's theorem now known as:

Chasles' theorem:

Any displacement of a rigid body is equivalent to the sum of a translation of any one point on that body and a rotation of the body about an axis through that point.

This describes complex motion as defined above and in Section 2.2 (p. 23). Note that equation 4.1c is an expression of Chasles' theorem.

4.4 GRAPHICAL POSITION ANALYSIS OF LINKAGES

For any one-*DOF* linkage, such as a fourbar, only one parameter is needed to completely define the positions of all the links. The parameter usually chosen is the angle of the input link. This is shown as θ_2 in Figure 4-4. We want to find θ_3 and θ_4. The link lengths are known. Note that we will consistently number the ground link as 1 and the driver link as 2 in these examples.

The graphical analysis of this problem is trivial and can be done using only high-school geometry. If we draw the linkage carefully to scale with rule, compass, and protractor in a particular position (given θ_2), then it is only necessary to measure the angles of links 3 and 4 with the protractor. Note that all link angles are measured from a positive X axis. In Figure 4-4, a *local xy* axis system, parallel to the *global XY* system, has been created at point A to measure θ_3. The accuracy of this graphical solution will be limited by our care and drafting ability and by the crudity of the protractor used. Nevertheless, a very rapid approximate solution can be found for any one position.

Figure 4-5 shows the construction of the graphical position solution. The four link lengths a, b, c, d and the angle θ_2 of the input link are given. First, the ground link (1) and the input link (2) are drawn to a convenient scale such that they intersect at the origin O_2 of the global XY coordinate system with link 2 placed at the input angle θ_2. Link 1 is drawn along the X axis for convenience. The compass is set to the scaled length of link 3, and an arc of that radius swung about the end of link 2 (point A). Then the compass is set to the scaled length of link 4, and a second arc swung about the end of link 1

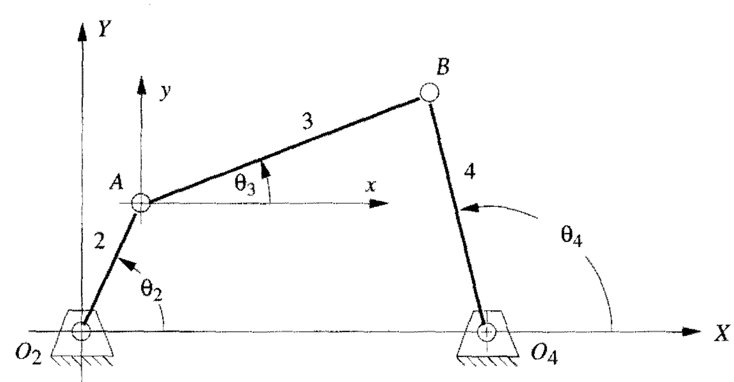


FIGURE 4-4
Measurement of angles in the fourbar linkage

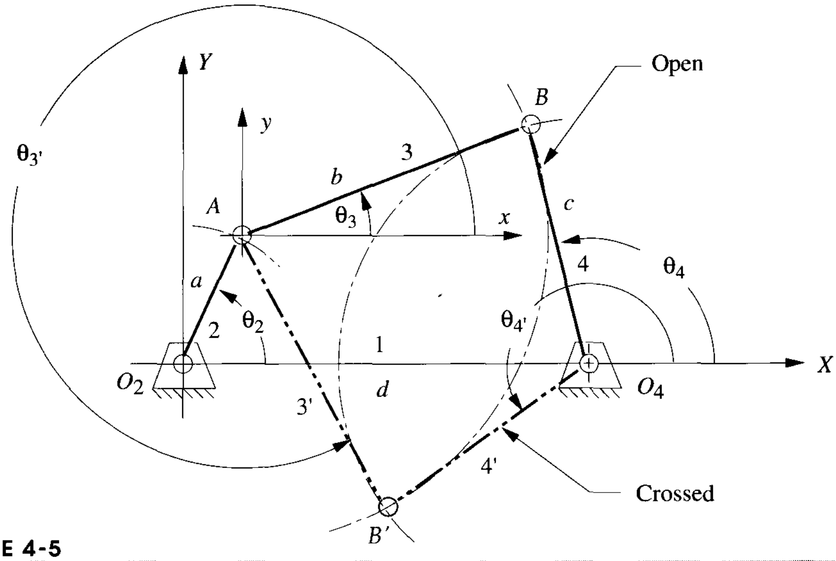


FIGURE 4-5

Graphical position solution to the open and crossed configurations of the fourbar linkage

(point O_4). These two arcs will have two intersections at B and B' that define the two solutions to the position problem for a fourbar linkage which can be assembled in two configurations, called circuits, labeled open and crossed in Figure 4-5. Circuits in linkages will be discussed in a later section.

The angles of links 3 and 4 can be measured with a protractor. One circuit has angles θ_3 and θ_4, the other $\theta_{3'}$ and $\theta_{4'}$. A graphical solution is only valid for the particular value of input angle used. For each additional position analysis we must completely redraw the linkage. This can become burdensome if we need a complete analysis at every 1- or 2-degree increment of θ_2. In that case we will be better off to derive an analytical solution for θ_3 and θ_4 which can be solved by computer.

4.5 ALGEBRAIC POSITION ANALYSIS OF LINKAGES

The same procedure that was used in Figure 4-5 to solve geometrically for the intersections B and B' and angles of links 3 and 4 can be encoded into an algebraic algorithm. The coordinates of point A are found from

$$A_x = a\cos\theta_2$$
$$A_y = a\sin\theta_2 \tag{4.2a}$$

The coordinates of point B are found using the equations of circles about A and O_4.

$$b^2 = (B_x - A_x)^2 + (B_y - A_y)^2 \tag{4.2b}$$

$$c^2 = (B_x - d)^2 + B_y^2 \tag{4.2c}$$

which provide a pair of simultaneous equations in B_x and B_y.

Subtracting equation 4.2c from 4.2b gives an expression for B_x.

$$B_x = \frac{a^2 - b^2 + c^2 - d^2}{2(A_x - d)} - \frac{2A_y B_y}{2(A_x - d)} = S - \frac{2A_y B_y}{2(A_x - d)} \tag{4.2d}$$

Substituting equation 4.2d into 4.2c gives a quadratic equation in B_y which has two solutions corresponding to those in Figure 4-5.

$$B_y^2 + \left(S - \frac{A_y B_y}{A_x - d} - d \right)^2 - c^2 = 0 \tag{4.2e}$$

This can be solved with the familiar expression for the roots of a quadratic equation,

$$B_y = \frac{-Q \pm \sqrt{Q^2 - 4PR}}{2P} \tag{4.2f}$$

where :

$$P = \frac{A_y^2}{(A_x - d)^2} + 1 \qquad Q = \frac{2A_y(d - S)}{A_x - d}$$

$$R = (d - S)^2 - c^2 \qquad S = \frac{a^2 - b^2 + c^2 - d^2}{2(A_x - d)}$$

Note that the solutions to this equation set can be real or imaginary. If the latter, it indicates that the links cannot connect at the given input angle or at all. Once the two values of B_y are found (if real), they can be substituted into equation 4.2d to find their corresponding x components. The link angles for this position can then be found from

$$\begin{aligned} \theta_3 &= \tan^{-1}\left(\frac{B_y - A_y}{B_x - A_x} \right) \\ \theta_4 &= \tan^{-1}\left(\frac{B_y}{B_x - d} \right) \end{aligned} \tag{4.2g}$$

A two-argument arctangent function must be used to solve equations 4.2g since the angles can be in any quadrant. Equations 4.2 can be encoded in any computer language or equation solver, and the value of 82 varied over the linkage's usable range to find all corresponding values of the other two link angles.

Vector Loop Representation of linkages

An alternate approach to linkage position analysis creates a vector loop (or loops) around the linkage. This approach offers some advantages in the synthesis of linkages which will be addressed in Chapter 5. The links are represented as **position** vectors. Figure 4-6 shows the same fourbar linkage as in Figure 4-4 (p. 151) , but the links are now drawn as position vectors which form a vector loop. This loop closes on itself making the sum of the vectors around the loop zero. The lengths of the vectors are the link lengths which

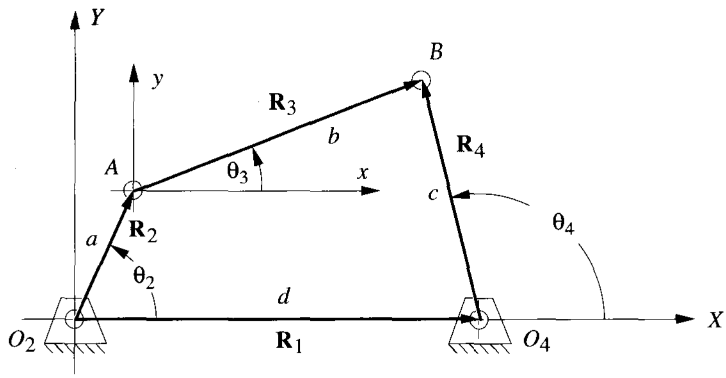


FIGURE 4-6

Position vector loop for a fourbar linkage

are known. The current linkage position is defined by the input angle θ_2 as it is a one-*DOF* mechanism. We want to solve for the unknown angles θ_3 and θ_4. To do so we need a convenient notation to represent the vectors.

Complex Numbers as Vectors

There are many ways to represent vectors. They may be defined in **polar coordinates**, by their *magnitude* and *angle*, or in **cartesian coordinates** as x and y components. These forms are of course easily convertible from one to the other using equations 4.0. The position vectors in Figure 4-6 can be represented as any of these expressions:

Polar form	Cartesian form	
$R @ \angle\theta$	$r\cos\theta\,\hat{\mathbf{i}} + r\sin\theta\,\hat{\mathbf{j}}$	(4.3a)
$re^{j\theta}$	$r\cos\theta + jr\sin\theta$	(4.3b)

Equation 4.3a uses **unit vectors** to represent the x and y vector component directions in the cartesian form. Figure 4-7 shows the unit vector notation for a position vector. Equation 4.3b uses **complex number notation** wherein the X direction component is called the *real portion* and the Y direction component is called the *imaginary portion.* This unfortunate term *imaginary* comes about because of the use of the notation j to represent the square root of minus one, which of course cannot be evaluated numerically. However, this *imaginary* number is used in a **complex number** as an **operator**, *not as a value.* Figure 4-8a shows the **complex plane** in which the *real* axis represents the X-directed component of the vector in the plane, and the *imaginary* axis represents the Y-directed component of the same vector. So, any term in a complex number which has no j operator is an x component, and a j indicates a y component.

Note in Figure 4-8b that each multiplication of the vector $\mathbf{R}_A$ by the operator j results in a *counterclockwise rotation* of the vector through 90 degrees. The vector $\mathbf{R}_B = j\mathbf{R}_A$ is directed along the *positive imaginary* or j axis. The vector $\mathbf{R}_C = j^2\,\mathbf{R}_A$ is directed along the *negative real* axis because $j^2 = -1$ and thus $\mathbf{R}_C = -\mathbf{R}_A$. In similar fashion, $\mathbf{R}_D = j^3\,\mathbf{R}_A = -j\mathbf{R}_A$ and this component is directed along the *negative j axis.*

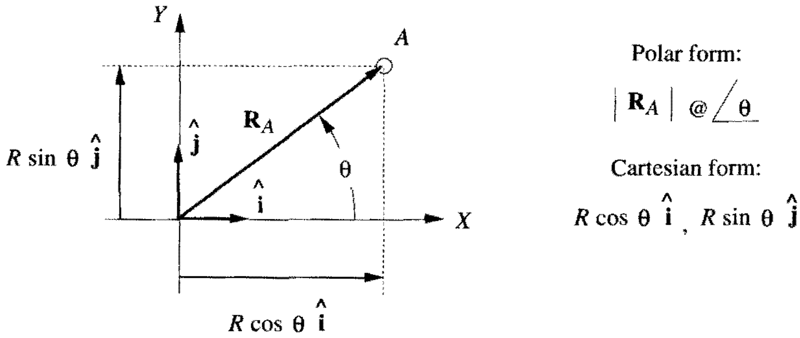


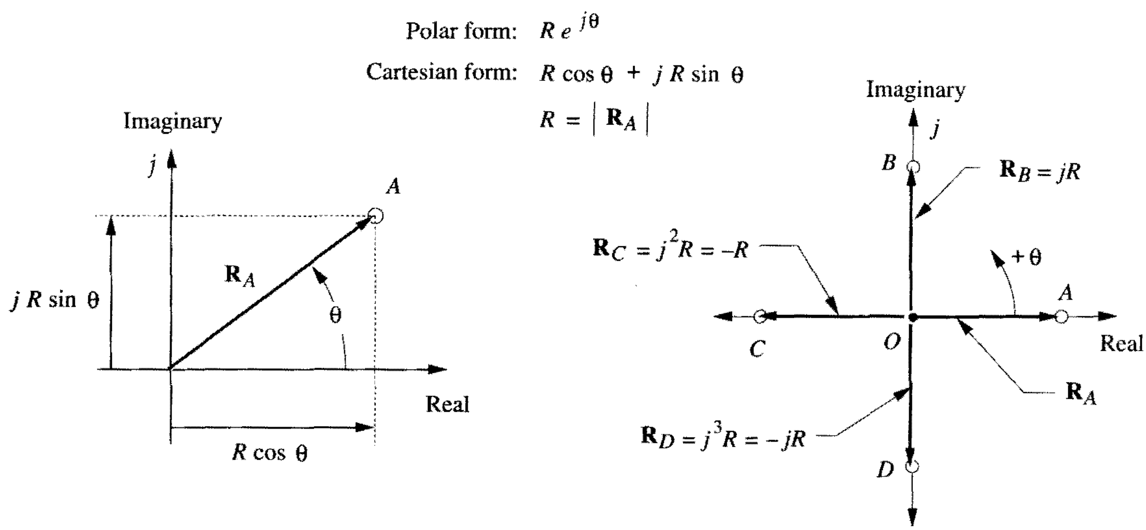
FIGURE 4-7

Unit vector notation for position vectors

One advantage of using this complex number notation to represent planar vectors comes from the **Euler identity**:

$$e^{\pm j\theta} = \cos\theta \pm j\sin\theta \tag{4.4a}$$

Any two-dimensional vector can be represented by the compact polar notation on the left side of equation 4.4a. There is no easier function to differentiate or integrate, since it is its own derivative:



(*a*) Complex number representation of a position vector (*b*) Vector rotations in the complex plane

FIGURE 4-8

Complex number representation of vectors in the plane

$$\frac{de^{j\theta}}{d\theta} = je^{j\theta} \tag{4.4b}$$

We will use this **complex number notation** for vectors to develop and derive the equations for position, velocity, and acceleration of linkages.

The Vector Loop Equation for a Fourbar Linkage

The directions of the position vectors in Figure 4-6 are chosen so as to define their angles where we desire them to be measured. By definition, *the angle of a vector is always measured at its root, **not at its head**.* We would like angle θ_4 to be measured at the fixed pivot O_4, so vector $\mathbf{R}_4$ is arranged to have its root at that point. We would like to measure angle θ_3 at the point where links 2 and 3 join, so vector $\mathbf{R}_3$ is rooted there. A similar logic dictates the arrangement of vectors $\mathbf{R}_1$ and $\mathbf{R}_2$. Note that the X (*real*) axis is taken for convenience along link 1 and the origin of the global coordinate system is taken at point O_2, the root of the input link vector, $\mathbf{R}_2$. These choices of vector directions and senses, as indicated by their arrowheads, lead to this vector loop equation:

$$\mathbf{R}_2 + \mathbf{R}_3 - \mathbf{R}_4 - \mathbf{R}_1 = 0 \tag{4.5a}$$

An **alternate notation** for these position vectors is to use the labels of the points at the vector **tips** and **roots** (*in that order*) as subscripts. The second subscript is conventionally omitted if it is the origin of the global coordinate system (point O_2):

$$\mathbf{R}_A + \mathbf{R}_{BA} - \mathbf{R}_{BO_4} - \mathbf{R}_{O_4} = 0 \tag{4.5b}$$

Next, we substitute the complex number notation for each position vector. To simplify the notation and minimize the use of subscripts, we will denote the scalar lengths of the four links as a, b, c, and d. These are so labeled in Figure 4-6. The equation then becomes:

$$ae^{j\theta_2} + be^{j\theta_3} - ce^{j\theta_4} - de^{j\theta_1} = 0 \tag{4.5c}$$

These are three forms of the same vector equation, and as such can be solved for two unknowns. There are four variables in this equation, namely the four link angles. The link lengths are all constant in this particular linkage. Also, the value of the angle of link 1 is fixed (at zero) since this is the ground link. The *independent variable* is θ_2 which we will control with a motor or other driver device. That leaves the angles of link 3 and 4 to be found. We need algebraic expressions which define θ_3 and θ_4 as functions only of the constant link lengths and the one input angle, θ_2. These expressions will be of the form:

$$\begin{aligned} \theta_3 &= f\{a, b, c, d, \theta_2\} \\ \theta_4 &= g\{a, b, c, d, \theta_2\} \end{aligned} \tag{4.5d}$$

To solve the polar form, vector equation 4.5c, we must substitute the *Euler equivalents* (equation 4.4a) for the $e\,j\theta$ terms, and then separate the resulting cartesian form vector equation into two scalar equations which can be solved simultaneously for θ_3 and θ_4. Substituting equation 4.4a into equation 4.5c:

$$a(\cos\theta_2 + j\sin\theta_2) + b(\cos\theta_3 + j\sin\theta_3) - c(\cos\theta_4 + j\sin\theta_4) - d(\cos\theta_1 + j\sin\theta_1) = 0 \tag{4.5e}$$

This equation can now be separated into its real and imaginary parts and each set to zero.

real part (x component):

$$a\cos\theta_2 + b\cos\theta_3 - c\cos\theta_4 - d\cos\theta_1 = 0$$

but: $\theta_1 = 0$, so: (4.6a)

$$a\cos\theta_2 + b\cos\theta_3 - c\cos\theta_4 - d = 0$$

imaginary part (y component):

$$ja\sin\theta_2 + jb\sin\theta_3 - jc\sin\theta_4 - jd\sin\theta_1 = 0$$

but: $\theta_1 = 0$, and the j's divide out, so: (4.6b)

$$a\sin\theta_2 + b\sin\theta_3 - c\sin\theta_4 = 0$$

The scalar equations 4.6a and 4.6b can now be solved simultaneously for θ_3 and θ_4. To solve this set of two simultaneous trigonometric equations is straightforward but tedious. Some substitution of trigonometric identities will simplify the expressions. The first step is to rewrite equations 4.6a and 4.6b so as to isolate one of the two unknowns on the left side. We will isolate θ_3 and solve for θ_4 in this example.

$$b\cos\theta_3 = -a\cos\theta_2 + c\cos\theta_4 + d \tag{4.6c}$$

$$b\sin\theta_3 = -a\sin\theta_2 + c\sin\theta_4 \tag{4.6d}$$

Now square both sides of equations 4.6c and 4.6d and add them:

$$b^2\left(\sin^2\theta_3 + \cos^2\theta_3\right) = \left(-a\sin\theta_2 + c\sin\theta_4\right)^2 + \left(-a\cos\theta_2 + c\cos\theta_4 + d\right)^2 \tag{4.7a}$$

Note that the quantity in parentheses on the left side is equal to 1, eliminating θ_3 from the equation, leaving only θ_4 which can now be solved for.

$$b^2 = \left(-a\sin\theta_2 + c\sin\theta_4\right)^2 + \left(-a\cos\theta_2 + c\cos\theta_4 + d\right)^2 \tag{4.7b}$$

The right side of this expression must now be expanded and terms collected.

$$b^2 = a^2 + c^2 + d^2 - 2ad\cos\theta_2 + 2cd\cos\theta_4 - 2ac\left(\sin\theta_2\sin\theta_4 + \cos\theta_2\cos\theta_4\right) \tag{4.7c}$$

To further simplify this expression, the constants K_1, K_2, and K_3 are defined in terms of the constant link lengths in equation 4.7c:

$$K_1 = \frac{d}{a} \qquad K_2 = \frac{d}{c} \qquad K_3 = \frac{a^2 - b^2 + c^2 + d^2}{2ac} \tag{4.8a}$$

and:

$$K_1\cos\theta_4 - K_2\cos\theta_2 + K_3 = \cos\theta_2\cos\theta_4 + \sin\theta_2\sin\theta_4 \tag{4.8b}$$

If we substitute the identity $\cos(\theta_2 - \theta_4) = \cos\theta_2\cos\theta_4 + \sin\theta_2\sin\theta_4$, we get the form known as Freudenstein's equation.

$$K_1\cos\theta_4 - K_2\cos\theta_2 + K_3 = \cos\left(\theta_2 - \theta_4\right) \tag{4.8c}$$

In order to reduce equation 4.8b to a more tractable form for solution, it will be useful to substitute the *half angle identities* which will convert the sin θ_4 and cos θ_4 terms to tan θ_4 terms:

$$\sin\theta_4 = \frac{2\tan\left(\frac{\theta_4}{2}\right)}{1+\tan^2\left(\frac{\theta_4}{2}\right)}; \qquad \cos\theta_4 = \frac{1-\tan^2\left(\frac{\theta_4}{2}\right)}{1+\tan^2\left(\frac{\theta_4}{2}\right)} \tag{4.9}$$

This results in the following simplified form, where the link lengths and known input value (θ_2) terms have been collected as constants A, B, and C.

$$A\tan^2\left(\frac{\theta_4}{2}\right) + B\tan\left(\frac{\theta_4}{2}\right) + C = 0$$

where:

$$\begin{aligned} A &= \cos\theta_2 - K_1 - K_2\cos\theta_2 + K_3 \\ B &= -2\sin\theta_2 \\ C &= K_1 - (K_2+1)\cos\theta_2 + K_3 \end{aligned} \tag{4.10a}$$

Note that equation 4.10a is quadratic in form, and the solution is:

$$\tan\left(\frac{\theta_4}{2}\right) = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

$$\theta_{4_{1,2}} = 2\arctan\left(\frac{-B \pm \sqrt{B^2 - 4AC}}{2A}\right) \tag{4.10b}$$

Equation 4.10b has two solutions, obtained from the ± conditions on the radical. These two solutions, as with any quadratic equation, may be of three types: *real and equal, real and unequal, complex conjugate.* If the discriminant under the radical is negative, then the solution is complex conjugate, which simply means that the link lengths chosen are not capable of connection for the chosen value of the input angle θ_2. This can occur either when the link lengths are completely incapable of connection in any position or, in a non-Grashof linkage, when the input angle is beyond a toggle limit position. There is then no real solution for that value of input angle θ_2. Excepting this situation, the solution will usually be real and unequal, meaning there are two values of θ_4 corresponding to any one value of θ_2. These are referred to as the **crossed** and **open** configurations of the linkage and also as the two **circuits** of the linkage. In the fourbar linkage, the minus solution gives θ_4 for the open configuration and the positive solution gives θ_4 for the crossed configuration.

Figure 4-5 (p. 152) shows both crossed and open solutions for a Grashof crank-rocker linkage. The terms crossed and open are based on the assumption that the input link 2, for which θ_2 is defined, is placed in the first quadrant (i.e., $0 < \theta_2 < \pi/2$). A Grashof linkage is then defined as **crossed** if the two links adjacent to the shortest link cross one another, and as **open** if they do not cross one another in this position. Note that the configuration of the linkage, either crossed or open, is solely dependent upon the way that the links are assembled. You cannot predict, based on link lengths alone, which of the solutions will be the desired one. In other words, you can obtain either solution with the same linkage by simply taking apart the pin which connects links 3 and 4 in Figure 4-5, and moving those links to the only other positions at which the pin will again connect them. In so doing, you will have switched from one position solution, or **circuit**, to the other.

The solution for angle θ_3 is essentially similar to that for θ_4. Returning to equations 4.6, we can rearrange them to isolate θ_4 on the left side.

$$c\cos\theta_4 = a\cos\theta_2 + b\cos\theta_3 - d \tag{4.6e}$$

$$c\sin\theta_4 = a\sin\theta_2 + b\sin\theta_3 \tag{4.6f}$$

Squaring and adding these equations will eliminate θ_4. The resulting equation can be solved for θ_3 as was done above for θ_4, yielding this expression:

$$K_1\cos\theta_3 + K_4\cos\theta_2 + K_5 = \cos\theta_2\cos\theta_3 + \sin\theta_2\sin\theta_3 \tag{4.11a}$$

The constant K_1 is the same as defined in equation 4.8b. K_4 and K_5 are:

$$K_4 = \frac{d}{b}; \qquad K_5 = \frac{c^2 - d^2 - a^2 - b^2}{2ab} \tag{4.11b}$$

This also reduces to a quadratic form:

$$D\tan^2\left(\frac{\theta_3}{2}\right) + E\tan\left(\frac{\theta_3}{2}\right) + F = 0$$

$$\text{where:} \quad \begin{aligned} D &= \cos\theta_2 - K_1 + K_4\cos\theta_2 + K_5 \\ E &= -2\sin\theta_2 \\ F &= K_1 + (K_4 - 1)\cos\theta_2 + K_5 \end{aligned} \tag{4.12}$$

and the solution is:

$$\theta_{3_{1,2}} = 2\arctan\left(\frac{-E \pm \sqrt{E^2 - 4DF}}{2D}\right) \tag{4.13}$$

As with the angle θ_4, this also has two solutions, corresponding to the crossed and open circuits of the linkage, as shown in Figure 4-5.

4.6 THE FOURBAR SLIDER-CRANK POSITION SOLUTION

The same vector loop approach as used above can be applied to a linkage containing sliders. Figure 4-9 shows an offset fourbar slider-crank linkage, inversion #1. The term **offset** means that *the slider axis extended does not pass through the crank pivot.* This is the general case. (The nonoffset slider-crank linkages shown in Figure 2-13 (p. 44) are the special cases.) This linkage could be represented by only three position vectors, $\mathbf{R}_2$, $\mathbf{R}_3$, and $\mathbf{R}_s$, but one of them ($\mathbf{R}_s$) will be a vector of varying magnitude and angle. It will be easier to use four vectors, $\mathbf{R}_1$, $\mathbf{R}_2$, $\mathbf{R}_3$, and $\mathbf{R}_4$ with $\mathbf{R}_1$ arranged parallel to the axis of slid-

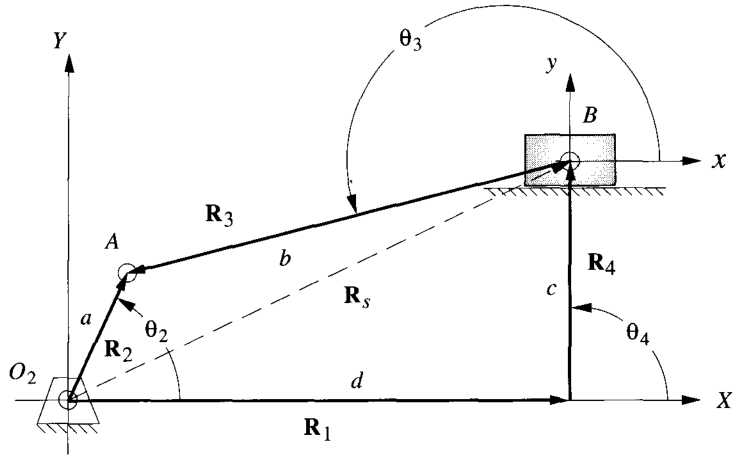


FIGURE 4-9
Position vector loop for a fourbar slider-crank linkage

ing and $\mathbf{R}_4$ perpendicular. In effect the pair of vectors $\mathbf{R}_1$ and $\mathbf{R}_4$ are orthogonal components of the position vector $\mathbf{R}_s$ from the origin to the slider.

It simplifies the analysis to arrange one coordinate axis parallel to the axis of sliding. The variable-length, constant-direction vector $\mathbf{R}_1$ then represents the slider position with magnitude d. The vector $\mathbf{R}_4$ is orthogonal to $\mathbf{R}_1$ and defines the constant magnitude **offset** of the linkage. Note that for the special-case, nonoffset version, the vector $\mathbf{R}_4$ will be zero and $\mathbf{R}_1 = \mathbf{R}_s$. The vectors $\mathbf{R}_2$ and $\mathbf{R}_3$ complete the vector loop. The coupler's position vector $\mathbf{R}_3$ is placed with its root at the slider which then defines its angle θ_3 at point B. This particular arrangement of position vectors leads to a vector loop equation similar to the pin-jointed fourbar example:

$$\mathbf{R}_2 - \mathbf{R}_3 - \mathbf{R}_4 - \mathbf{R}_1 = 0 \tag{4.14a}$$

Compare equation 4.14a to equation 4.5a (p. 156) and note that the only difference is the sign of $\mathbf{R}_3$. This is due solely to the somewhat arbitrary choice of the sense of the position vector $\mathbf{R}_3$ in each case. The angle θ_3 must always be measured at the root of vector $\mathbf{R}_3$, and in this example it will be convenient to have that angle θ_3 at the joint labeled B. Once these arbitrary choices are made it is crucial that the resulting algebraic signs be carefully observed in the equations, or the results will be completely erroneous. Letting the vector magnitudes (link lengths) be represented by a, b, c, d as shown, we can substitute the complex number equivalents for the position vectors.

$$a e^{j\theta_2} - b e^{j\theta_3} - c e^{j\theta_4} - d e^{j\theta_1} = 0 \tag{4.14b}$$

Substitute the Euler equivalents:

$$\begin{aligned} &a(\cos\theta_2 + j\sin\theta_2) - b(\cos\theta_3 + j\sin\theta_3) \\ &\quad - c(\cos\theta_4 + j\sin\theta_4) - d(\cos\theta_1 + j\sin\theta_1) = 0 \end{aligned} \tag{4.14c}$$

Separate the real and imaginary components:

real part (x component):

$$a\cos\theta_2 - b\cos\theta_3 - c\cos\theta_4 - d\cos\theta_1 = 0$$

$$\text{but: } \theta_1 = 0, \text{ so:} \tag{4.15a}$$

$$a\cos\theta_2 - b\cos\theta_3 - c\cos\theta_4 - d = 0$$

imaginary part (y component):

$$ja\sin\theta_2 - jb\sin\theta_3 - jc\sin\theta_4 - jd\sin\theta_1 = 0$$

$$\text{but: } \theta_1 = 0, \text{and the } j\text{'s divide out, so:} \tag{4.15b}$$

$$a\sin\theta_2 - b\sin\theta_3 - c\sin\theta_4 = 0$$

We want to solve equations 4.15 simultaneously for the two unknowns, link length d and link angle θ_3. The independent variable is crank angle θ_2. Link lengths a and b, the offset c, and angle θ_4 are known. But note that since we set up the coordinate system to be parallel and perpendicular to the axis of the slider block, the angle θ_1 is zero and θ_4 is 90°. Equation 4.15b can be solved for θ_3 and the result substituted into equation 4.15a to solve for d. The solution is:

$$\theta_{3_1} = \arcsin\left(\frac{a\sin\theta_2 - c}{b}\right) \tag{4.16a}$$

$$d = a\cos\theta_2 - b\cos\theta_3 \tag{4.16b}$$

Note that there are again two valid solutions corresponding to the two circuits of the linkage. The arcsine function is multivalued. Its evaluation will give a value between ±90° representing only one circuit of the linkage. The value of d is dependent on the calculated value of θ_3. The value of θ_3 for the second circuit of the linkage can be found from:

$$\theta_{3_2} = \arcsin\left(-\frac{a\sin\theta_2 - c}{b}\right) + \pi \tag{4.17}$$

4.7 AN INVERTED SLIDER-CRANK POSITION SOLUTION

Figure 4-l0a shows inversion #3 of the common fourbar slider-crank linkage in which the sliding joint is between links 3 and 4 at point *B*. This is shown as an offset slider-crank mechanism. The slider block has pure rotation with its center offset from the slide axis. (Figure 2-l3c, p. 44, shows the nonoffset version of this linkage in which the vector R.t is zero.)

The global coordinate system is again taken with its origin at input crank pivot 02 and the positive X axis along link 1, the ground link. A local axis system has been placed at point *B* in order to define 83. Note that there is a fixed angle 'Ywithin link 4 which defines the slot angle with respect to that link.

In Figure 4-l0b the links have been represented as position vectors having senses consistent with the coordinate systems that were chosen for convenience in defining the

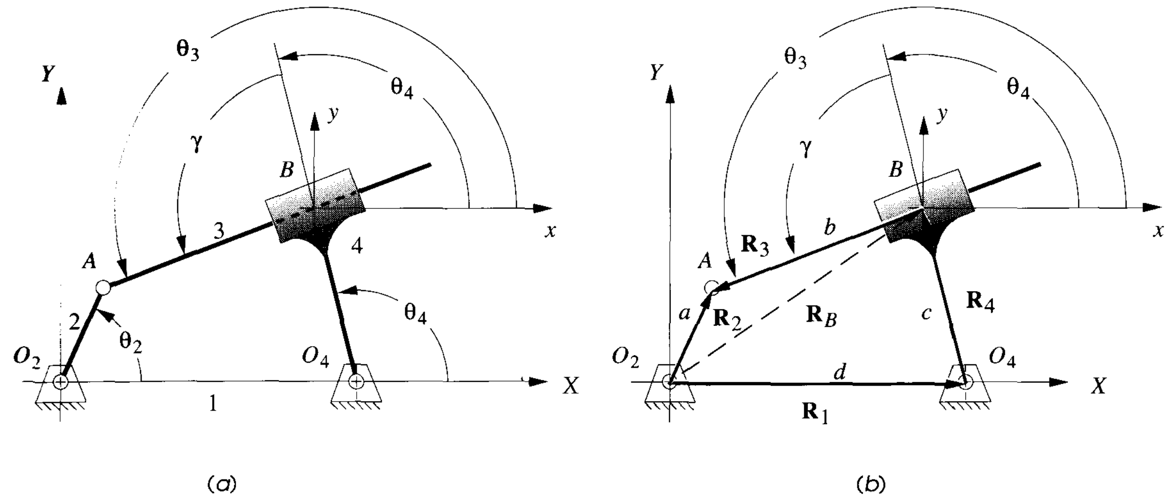


FIGURE 4-10
Inversion #3 of the slider-crank fourbar linkage

link angles. This particular arrangement of position vectors leads to the same vector loop equation as the previous slider-crank example. Equations 4.14 and 4.15 apply to this inversion as well. Note that the absolute position of point B is defined by vector $\mathbf{R}_B$ which varies in both magnitude and direction as the linkage moves. We choose to represent $\mathbf{R}_B$ as the vector difference $\mathbf{R}_2 - \mathbf{R}_3$ in order to use the actual links as the position vectors in the loop equation.

All slider linkages will have at least one link whose effective length between joints will vary as the linkage moves. In this example the length of link 3 between points A and B, designated as b, will change as it passes through the slider block on link 4. Thus the value of b will be one of the variables to be solved for in this inversion. Another variable will be θ_4, the angle of link 4. Note however, that we also have an unknown in θ_3, the angle of link 3. This is a total of three unknowns. Equations 4.15 can only be solved for two unknowns. Thus we require another equation to solve the system. There is a fixed relationship between angles θ_3 and θ_4, shown as γ in Figure 4-10, which gives the equation:

$$\theta_3 = \theta_4 \pm \gamma \tag{4.18}$$

where the + sign is used for the open and the – sign for the crossed configuration.

Substituting equation 4.18 into equations 4.15 yields:

$$a\cos\theta_2 - b\cos\theta_3 - c\cos\theta_4 - d = 0 \tag{4.19a}$$
$$a\sin\theta_2 - b\sin\theta_3 - c\sin\theta_4 = 0 \tag{4.19b}$$

These have only two unknowns and can be solved simultaneously for θ_4 and b. Equation 4.19b can be solved for link length b and substituted into equation 4.19a.

$$b = \frac{a\sin\theta_2 - c\sin\theta_4}{\sin\theta_3} \tag{4.20a}$$

$$a\cos\theta_2 - \frac{a\sin\theta_2 - c\sin\theta_4}{\sin\theta_3}\cos\theta_3 - c\cos\theta_4 - d = 0 \tag{4.20b}$$

Substitute equation 4.18 and after some algebraic manipulation, equation 4.20 can be reduced to:

$$P\sin\theta_4 + Q\cos\theta_4 + R = 0$$

where: (4.21)

$$P = a\sin\theta_2\sin\gamma + (a\cos\theta_2 - d)\cos\gamma$$
$$Q = -a\sin\theta_2\cos\gamma + (a\cos\theta_2 - d)\sin\gamma$$
$$R = -c\sin\gamma$$

Note that the factors P, Q, R are constant for any input value of θ_2. To solve this for θ_4, it is convenient to substitute the tangent half angle identities (equation 4.9, p.158) for the sin θ_4 and cos θ_4 terms. This will result in a quadratic equation in tan (θ_4 / 2) which can be solved for the two values of θ_4.

$$P\frac{2\tan\left(\frac{\theta_4}{2}\right)}{1+\tan^2\left(\frac{\theta_4}{2}\right)} + Q\frac{1-\tan^2\left(\frac{\theta_4}{2}\right)}{1+\tan^2\left(\frac{\theta_4}{2}\right)} + R = 0 \tag{4.22a}$$

This reduces to:

$$(R-Q)\tan^2\left(\frac{\theta_4}{2}\right) + 2P\tan\left(\frac{\theta_4}{2}\right) + (Q+R) = 0$$

let:

$$S = R - Q; \qquad T = 2P; \qquad U = Q + R$$

then:

$$S\tan^2\left(\frac{\theta_4}{2}\right) + T\tan\left(\frac{\theta_4}{2}\right) + U = 0 \tag{4.22b}$$

and the solution is:

$$\theta_{4_{1,2}} = 2\arctan\left(\frac{-T \pm \sqrt{T^2 - 4SU}}{2S}\right) \tag{4.22c}$$

As was the case with the previous examples, this also has a crossed and an open solution represented by the plus and minus signs on the radical. Note that we must also calculate the values of link length b for each θ_4 by using equation 4.20a. The coupler angle θ_3 is found from equation 4.18.